Physics 468/668 (Spring 2018)
Thermodynamics/Statistical Physics
Mo/We 1:00-2:15 p.m., BPB 250

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Office hrs	Mo/We 2:15-3:15 p.m.

Course Outline:

- Statistical basis of thermodynamics
- Ensemble theory
- (Micro) canonical ensemble
- Grand canonical ensemble
- Phase transitions

Prerequisite: PHY 182, PHY 467
Credit Hours: 3
Textbook: *Statistical Mechanics*, R. K. Pathria and P. D. Beale

Grade Distribution:

Assignments	30%
Midterm Exam	30%
Final Exam	40%
Extra Credits	10%

Course Description:

This course covers elements of statistical physics from fundamental of classical statistic mechanics to quantum statistics. Approximately, we spend two weeks for each chapter (1-7 in Pathria's textbook). The weekly coverage might change as it depends on the progress of the class. There will be one exam during the semester, either take-home or in-class work. This final exam will cover all materials taught in this semester. You may work with others on the homework, but take-home exams must be done **strictly by yourself**. Barring documentable emergencies or observance of a certifiable religious holiday, all exams must be taken at the time and place specified.

Learning Outcomes:

- understand the microscopic basis for thermodynamics
- understand the macroscopic properties from micro level
- understand phase transition

Physics 468/668: Statistical Physics **Spring 2019**

Lecture 1: From Thermodynamics to Statistics

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

1.1 An overview about thermodynamics

Thermodynamics is aimed at describing the macroscopic phenomenon such as heat transfer, energy conversion, chemical reactions. We study them by measuring some key quantities (such as pressure, temperature, volume, viscosity, bulk modulus, .etc) and their correlations. The foundation of thermodynamics is based on the thermodynamic laws:

1. energy cannot be created or destroyed in an isolated system.
2. the entropy of any isolated system always increases
3. the entropy of a system approaches a constant value as the temperature approaches absolute zero

We can understand the general picture by studying the following equations

$$PV = nRT = NkT \tag{1.1}$$

$$N = n \times N_A \tag{1.2}$$

$$U_{thermal} = N \cdot f \cdot \frac{1}{2}kT \tag{1.3}$$

$$\Delta U = Q + W \tag{1.4}$$

$$\Delta W = -P\Delta V \tag{1.5}$$

$$S = k\ln\Omega \tag{1.6}$$

$$dU = TdS - PdV + \mu dN \tag{1.7}$$

However, thermodynamics does not tell us the microscopic details about the matter. We observe heat always flows spontaneously from a hot object to a cold one. The underlying explanation of this thermodynamic phenomenon is due to the change of entropy, a microscopic phenomenon. We might briefly discussed them to some extents in thermodynamic classes. Here, we are going to to explore them more comprehensively with the tool of statistic mechanics.

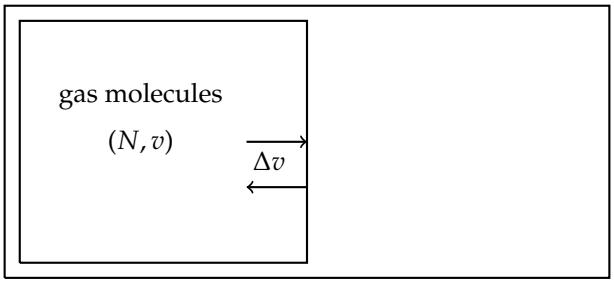


Figure 1.1: A schematic of gases in a container.

1.2 Kinetic Theory of Gases

In the early days of thermal physics, it was primarily shaped by experimental studies of the macroscopic behavior of physical systems, through the work of Carnot, Joule, Clausius and Kelvin. However, people were keen to know how the observed phenomenon were related to the motion of molecules and atoms.

The first attempt was the so called kinetic theory of gases, which aimed at explaining the macroscopic behavior of gaseous systems in terms of the motion of molecules. Although a bit speculative, it began to emerge as a real mathematical theory.

The theory for ideal gases makes the following assumptions:

1. The gas consists of very small particles known as molecules separate by large distances
2. The number of molecules is so large that statistical treatment can be applied.
3. These molecules are in constant, random, and rapid motion.
4. The rapidly moving particles constantly collide among themselves and with the walls of the container.
5. All these collisions are perfectly elastic.
6. Except during collisions, the interactions among molecules are negligible.

With these assumptions, we then proceed to figure the relations between the pressure and atomic motion. In such model, the pressure is equal to the force exerted by the atoms hitting and rebounding from a unit area of wall. Consider a gas of N molecules, each of mass m, enclosed in a cube with $V = L^3$. When a gas molecule collides with the wall perpendicular to the x axis and bounces off in the opposite direction with the same speed (an elastic collision), the change in momentum is given by:

$$\Delta P = p_{i,x} - p_{f,x} = m(v_x) - m(-v_x) = 2mv_x \tag{1.8}$$

where p is the momentum, i and f indicate the initial and final state, v is the speed of particle. The particle impacts once wall once every

$$\Delta t = \frac{2L}{v_x} \tag{1.9}$$

where L is the distance between two walls. The force contributed from one particle can be expressed as

$$F = \frac{\Delta p}{\Delta t} = \frac{mv_x^2}{L} \tag{1.10}$$

Since we have many particles, the total force on the wall is:

$$F = \frac{m\bar{v_x^2}}{L} \tag{1.11}$$

Since the motion of the particles is random and there is no bias applied to any direction, the averaged speed in each direction should be identical.

$$\bar{v_x^2} = \bar{v_y^2} = \bar{v_z^2} \tag{1.12}$$

$$\bar{v^2} = \bar{v_x^2} + \bar{v_y^2} + \bar{v_z^2} = 3\bar{v_x^2} \tag{1.13}$$

And the force is

$$F = \frac{Nm\bar{v^2}}{2L} \tag{1.14}$$

So we obtain the pressure

$$P = \frac{F}{L^2} = \frac{Nm\bar{v^2}}{3V} \tag{1.15}$$

In term of the kinetic energy ($K = 1/2Nm\bar{v^2}$),

$$PV = \frac{2K}{3} \tag{1.16}$$

This is a first non-trivial result of the kinetic theory because it relates pressure (a macroscopic property), to the (translational) kinetic energy of the molecules (a microscopic property).

From the ideal gas law $PV = nRT = NkT$, we can further derive the relation between K and T.

$$PV = \frac{2K}{3} = NkT \tag{1.17}$$

As one can see, this simple scheme could successfully explain some key features in an elegant way. However, it is limited by the oversimplified assumptions. The real contact with thermodynamics could not be made until 1872 when Boltzmann developed his H-theorem and established a direct connection between entropy and molecular dynamics. Simultaneously, the kinetic theory began giving away to its more sophisticated successor - the ensemble theory. The power of the techniques that finally emerged reduced thermodynamics to the status of an consequence of the statistics and the mechanics of the molecules constituting a given physical system. It was then natural to give the formalism name *Statistical Mechanics*

Physics 468/668: Statistical Physics **Spring 2019**

Lecture 2: The Statistical Basis of Thermodynamics

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

2.1 Microstates and Macrostates

We consider a system composed of N identical particles confined to a space of V. The total energy E would be equal to the sum of the energies of the individual particles.

$$E = \sum_i n_i \epsilon_i \tag{2.1}$$

The specification of the actual values of the parameters N, V, E then defines a *macrostate* of the system. At the molecular level, however, a large number of possibilities still exist because at that level there will be a large number of different ways to make the total state of N, V, E (think out arranging the coins with different sequence of head and tail). Each of the different ways specifies a *microstate* or *complexion* of the given system.

The actual number of all possible microstates (Ω) will be a function of N, V, E. In principle, it is from the magnitude of the number of Ω and from its dependence on the parameters N, V, E, that complete thermodynamics can be derived.

2.2 Multiplicity in Einstein Solids

N: Number of the oscillators.
q: Number of energy states.

$$\Omega(N, q) = \binom{q + N - 1}{q} = \frac{(q + N - 1)!}{q!(N - 1)!} \tag{2.2}$$

It can be simply proved as follows
q circles;
$N - 1$ vertical lines;
how to arrange them?



Exercises
Calculate the multiplicity of an Einstein solid with 5 oscillators and [1,2,3,4,5] units of Energy.

q	$\Omega(5,q)$
1	
2	
3	
4	
5	

2.3 Contact between statistics and thermodynamics

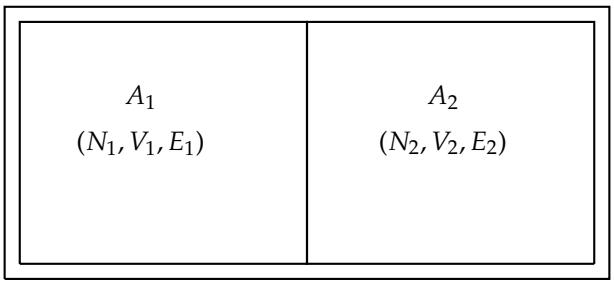


Figure 2.1: A schematic of two physical systems in thermal contact.

Let's first figure out how Ω is related to the thermodynamic quantities. We consider two physical systems, A_1 and A_2, which are separately in equilibrium. Let the macrostate of A_1 be represented by the parameters N_1, $V_1$and E_1 so that it has $\Omega_1(N_1, V_1, E_1)$ possible microstates, and the macrostate of A_2 be represented by $\Omega_2(N_2, V_2, E_2)$. Can we derive some thermodynamic properties from $\Omega_1(N_1, V_1, E_1)$ and $\Omega_2(N_2, V_2, E_2)$?

Let's bring two systems into thermal contact. For simplicity, we only allow the heat exchange between the two, while N, V remain fixed. This means there could be some interchanges between E_1 and E_2, however, it has to be restricted by the conservation law.

$$E = E_1 + E_2 = \text{const} \tag{2.3}$$

From the microscopic view, the total number of microstates could be expressed as,

$$\Omega_1(E_1)\Omega_2(E_2) = \Omega_1(E_1)\Omega_2(E - E_1) \tag{2.4}$$

When the system approaches to the equilibrium, what should be the value of $\bar{E_1}$? According to the 2nd law, the entropy should reach the maximum. Mathematically, we need to find $\bar{E_1}$ which satisfies,

$$\left(\frac{\partial\Omega_1(E_1)}{\partial E_1}\right)_{E_1=\bar{E_1}}\Omega_2(E_2) + \left(\frac{\partial\Omega_2(E_2)}{\partial E_2}\right)_{E_2=\bar{E_2}}\frac{\partial E_2}{\partial E_1}\Omega_1(E_1) = 0 \tag{2.5}$$

Remember that $\Delta E_1 = \Delta E_2$ at each time interval, therefore

$$\left(\frac{\partial \ln\Omega_1(E_1)}{\partial E_1}\right)_{E_1=\bar{E_1}} = \left(\frac{\partial \ln\Omega_2(E_2)}{\partial E_2}\right)_{E_2=\bar{E_2}} \tag{2.6}$$

Thus, our condition for equilibrium reduces to the equality of parameter β_1 and β_2:

$$\beta \equiv \left(\frac{\partial \ln\Omega(E)}{\partial E}\right)_{E=\bar{E}} \tag{2.7}$$

or a more complete version as follows,

$$\beta \equiv \left(\frac{\partial \ln\Omega(N,V,E)}{\partial E}\right)_{N,V,E=\bar{E}} \tag{2.8}$$

Therefore, we find when two systems are into thermal contact, the exchange of heat continues until the equilibrium E_1, E_2 reach some values. This happens only when the respective values of β_1 and β_2 become equal. It is then natural to expect that the parameter β is somehow related to T. To determine this relationship, we recall the thermodynamic formula

$$\left(\frac{\partial S}{\partial E}\right)_{N,V} = \frac{1}{T} \tag{2.9}$$

Comparing eq. 2.8 and 2.9, we find

$$\left(\frac{\Delta S}{\Delta \ln\Omega}\right) = \frac{1}{\beta T} = \text{const} \tag{2.10}$$

This correspondence was firstly established by Boltzmann. It was Planck who first wrote the explicit formula

$$S = k \ln\Omega \tag{2.11}$$

It means that the absolute value of the entropy of a given physical system in terms of the total number of microstates accessible to it conformity with the given macrostate, which provides a bridge between micro and macroscopic.

2.4 More complete contact

Let's continue to examine a more elaborate exchange between A_1 and A_2.

not only,

$$\left(\frac{\partial \ln\Omega_1(E_1)}{\partial E_1}\right)_{E_1=\bar{E_1}} = \left(\frac{\partial \ln\Omega_2(E_2)}{\partial E_2}\right)_{E_2=\bar{E_2}} \tag{2.12}$$

but also

$$\left(\frac{\partial \ln\Omega_1(V_1)}{\partial V_1}\right)_{V_1=\bar{V_1}} = \left(\frac{\partial \ln\Omega_2(V_2)}{\partial V_2}\right)_{V_2=\bar{V_2}} \tag{2.13}$$

Our conditions for equilibrium now take the form of an equality between the pair of (β, η)

$$\eta \equiv \left(\frac{\partial \ln\Omega(N,V,E)}{\partial V}\right)_{N,E,V=\bar{V}} \tag{2.14}$$

Similarly, there might be exchanges between particles, while need another parameter ζ,

$$\zeta \equiv \left(\frac{\partial \ln\Omega(N,V,E)}{\partial N}\right)_{V,E,N=\bar{N}} \tag{2.15}$$

To determine the physical meaning of the parameters η and ζ, we make use of the thermodynamic identity.

$$dE = TdS - PdV + \mu dN \tag{2.16}$$

so

$$\begin{aligned}\beta &= \frac{1}{kT} \\ \eta &= \frac{P}{kT} \\ \zeta &= -\frac{\mu}{kT}\end{aligned} \tag{2.17}$$

From the macroscopic view, the equilibrium is reached when

$$\begin{aligned}T_1 &= T_2 \\ P_1 &= P_2 \\ \mu_1 &= \mu_2\end{aligned} \tag{2.18}$$

This is identical to the ones following from statistical considerations. The evaluations of P, μ, T indeed requires that energy E be expressed as a function of N, V, E, this should, in principle be possible once S is known.

For instance,

$$\begin{aligned}\left(\frac{\partial S}{\partial E}\right)_{N,V} &= \frac{1}{T} \\ \left(\frac{\partial S}{\partial V}\right)_{N,V} &= \frac{P}{T} \\ \left(\frac{\partial S}{\partial N}\right)_{N,V} &= \frac{-\mu}{T}\end{aligned} \tag{2.19}$$

The rest of thermodynamic quantities follow straightforwardly.

$$\begin{aligned}F &= E - TS \\ G &= F + PV = E - TS - PV = \mu N \\ H &= E + PV = G + TS\end{aligned} \tag{2.20}$$

$$\begin{aligned}C_V &= T(\frac{\partial S}{\partial T})_{N,V} = (\frac{\partial E}{\partial T})_{N,V} \\ C_P &= T(\frac{\partial S}{\partial T})_{N,P} = (\frac{\partial H}{\partial T})_{N,P}\end{aligned} \tag{2.21}$$

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Lecture 3: The Statistical Basis of Thermodynamics

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

3.1 The classical ideal gas

The key to derive the thermodynamic quantities is to solve S. We shall now derive the various thermodynamic properties of a classical ideal gas composed of monoatomic molecules. The main reason why we choose this high specialized system for consideration is that it affords an explicit, asympototic evaluation of the number $\Omega(N, V, E)$. In addition, this study serves as a useful reference with which the behavior of other physical systems, especially real gases, can be compared.

Again, we shall start with some assumptions.

1. noninteracting particles

2. distinguishable

$$\begin{aligned} \text{One particle:} \quad & \Omega(1, E, V) \propto V \\ \text{N particles:} \quad & \Omega(N, E, V) \propto V^N \end{aligned} \tag{3.1}$$

$$\frac{P}{T} = k\left(\frac{\partial \ln \Omega(N, E, V)}{\partial V}\right)_{N,E} = k\frac{N}{V} \tag{3.2}$$

If the system contains n moles of the gas, then $N = nN_A$, where N_A is the Avogadro number. The above equation can lead to ideal gas law.

$$PV = NkT = nRT \tag{3.3}$$

3.2 Intensive and extensive properties

Physical properties of materials and systems can often be categorized as being either *intensive* or *extensive* quantities, according to how the property changes when the size (or extent) of the system changes.

1. An *intensive* property is one whose magnitude is independent of the size of the system. (P,T)

2. An *extensive* property is one whose magnitude is additive for subsystems. (S, V)

3.3 Compute the Multiplicity

To derive other quantities, we need a detailed knowledge of the way Ω depends on the parameters N, V, E. The problem essentially reduces to determining the total number of ways satisfying the equation.

$$\sum_{r=1}^{3N} \epsilon_r = E \tag{3.4}$$

where ϵ_r are the energies associated with the various degrees of freedom of the N particles. For a molecule confined to the box, its wave-function must go to zero at each end, and therefore the allowed standing wave patterns are limited to the wavelengths of

$$\lambda_n = \frac{2L}{n}, \quad n = 1, 2, 3, ... \tag{3.5}$$

Thus

$$p_n = \frac{h}{\lambda_n} = \frac{hn}{2L} \quad n = 1, 2, 3, ... \tag{3.6}$$

Now the energy eigenvalues for a free, non-relativistic particle confined to a cubic box of side L, under the condition that the wave-function $\Psi(r)$ vanishes everywhere on the boundary are given by

$$\epsilon_r(n_x, n_y, n_z) = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2); \quad n_x, n_y, n_z = 1, 2, 3, ... \tag{3.7}$$

The number of distinct eigenfunctions should be equal to the number of independent, positive-integral solutions of the equation

$$(n_x^2 + n_y^2 + n_z^2) = \frac{8mV^{2/3}\epsilon}{h^2} = \epsilon^* \tag{3.8}$$

We may denote the number by $\Omega(1, \epsilon, V)$. Extending this argument, it follows that the desired number $\Omega(N, E, V)$ would be equal to the number of independent, positive integral solutions of the equation.

$$\sum_{r=1}^{3N} n_r^2 = \frac{8mV^{2/3}\epsilon}{h^2} = E^* \tag{3.9}$$

From the expression on the right side, we conclude that the **volume** V **and the energy** E of the system enter into the expression for Ω in the form of the combination $(V^{2/3}E)$.

$$S(N, V, E) = S(N, V^{2/3}E). \tag{3.10}$$

Hence, for the **consistency of** S **and** N, which defines a **reversible adiabatic process**,

$$V^{2/3}E = \text{const} \tag{3.11}$$

$$P = -\left(\frac{\partial E}{\partial V}\right)_{N,S} = \frac{2E}{3V} \tag{3.12}$$

Also, we shall get,

$$PV^{5/3} = \text{const} \tag{3.13}$$

which tells us how P varies with V during the reversible adiabatic process.

Question: how the relation was derived in thermodynamic class?

3.4 From the ideal gas picture

Suppose we have a single gas atom (Ar), with kinetic energy U in a container of volume V, what is its corresponding Ω? Obviously, the possible microstate is proportional to V. In principle, the atom can stay at any place of V. Also, each microstate can be represented as a vector, since it has velocity (more precisely Momentum). Therefore

$$\Omega \approx V \cdot V_p \tag{3.14}$$

It appears that both V and V_p somehow relate to very large numbers, would their product go to infinity? Fortunately, we have the famous **Heisenberg uncertainty principle**:

$$\Delta x \Delta p_x = h. \tag{3.15}$$

For a one-dimensional chain, we define L as the length in real space, L_p as the length in momentum space,

$$\Omega_{1D} = \frac{L}{\Delta x}\frac{L_p}{\Delta p_x} = \frac{LL_p}{h}. \tag{3.16}$$

Therefore, its 3D version is,

$$\Omega_1 = \frac{VV_p}{h^3}. \tag{3.17}$$

Accordingly, the multiplicity function for an ideal gas of two molecules should be

$$\Omega_2 = \frac{1}{2}\frac{V^2}{h^6} \times \text{area of } P \text{ hypersphere} \tag{3.18}$$

if the two molecules are indistinguishable. The general form for N should be

$$\Omega_N = \frac{1}{N!}\frac{V^N}{h^{3N}} \times \text{area of } P \text{ hypersphere}. \tag{3.19}$$

For N=1, how to calculate the area? Since E depends on the momentum by

$$E = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2) = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) \tag{3.20}$$

$$p_x^2 + p_y^2 + p_z^2 = \sqrt{2mE} \tag{3.21}$$

So the momentum space is the surface of a sphere with radius $\sqrt{2mE}$, namely,

$$\begin{aligned}
\text{area} &= 2 && (d\text{=}1)\\
&= 2\pi r && (d\text{=}2)\\
&= 4\pi r^2 && (d\text{=}3)\\
&= .... &&\\
&= .... &&\\
&= \frac{2\pi^{d/2}}{\Gamma(d/2)} r^{d-1} && (d \text{ in general})
\end{aligned} \tag{3.22}$$

Therefore, the general Ω is

$$\Omega_N = \frac{1}{N!}\frac{V^N}{h^{3N}}\frac{2\pi^{3N/2}}{(3N/2-1)!}\sqrt{2mE}^{3N-1}. \tag{3.23}$$

$$\Omega(U,V,N) = f(N)V^N E^{3N/2} \tag{3.24}$$

where $f(N)$ is a complicated function of N.

$$\ln\Omega \approx \ln\Sigma \approx N\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}N \tag{3.25}$$

$$S(N,V,E) = k\ln\Omega = Nk\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}Nk \tag{3.26}$$

which can be inverted to give

$$E(S,V,N) = \frac{3h^2N}{4\pi mV^{2/3}}\exp(\frac{2S}{3Nk} - 1) \tag{3.27}$$

The temperature of the gas then follows:

$$T = \frac{\partial E}{\partial S} = \frac{2}{3Nk}E \tag{3.28}$$

$$E = N(\frac{3}{2}kT) \tag{3.29}$$

$$C_V = (\frac{\partial E}{\partial T})_{N,V} = \frac{3}{2}Nk \tag{3.30}$$

$$C_P = (\frac{\partial E + PV}{\partial T})_{N,V} = \frac{5}{2}Nk \tag{3.31}$$

$$P = (\frac{\partial E}{\partial V})_{N,S} = \frac{2E}{3V} \tag{3.32}$$

The ratio between C_P and C_V is $\gamma = C_P/C_V = 5/3$.

The considerations of this section clearly demonstrated that the thermodynamics of a macroscopic system can be derived from the Ω. The whole problem then hinges on an asymptotic enumeration of these numbers, which unfortunately is tractable only for a few cases. But there remains an inadequacy that could not be detected in the derivations made so far, this relates to the explicit dependence of S and N.

3.5 Homework

1. prove $PV^{5/3}$ is a constant for ideal gas under adiabatic process.
2. 1.1a (do Taylor expansion for $\Omega(E)$ on $E - \bar{E}$)
3. 1.3 ($dS >= 0$)
4. 1.7 (see lecture)
5. 1.9 (express $S(N,V,E)$ as $Nf(V/N, E/N)$)

Physics 468/668: Statistical Physics **Spring 2019**

Lecture 4: The Statistical Basis of Thermodynamics

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

4.1 The entropy of mixing and Gibbs paradox

Recall that we finally derived the expression for the entropy of ideal gas in the previous lecture, with a couple of assumptions.

1. noninteracting particles
2. distinguishable

$$S(N,V,E) = Nk\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}Nk \tag{4.1}$$

If we really dig into the equation, we readily observe that the entropy is not an extensive property of the system! That is, if we increase the size of the system by a factor α, keeping the intensive variables unchanged, then the entropy of the system, which should also increase by the same factor, does not do so. This is due to the presence of $\ln V$ term. The conclusion here simply means that the entropy of this system is different from the sum of the entropies of its parts. This is known as *Gibbs paradox*.

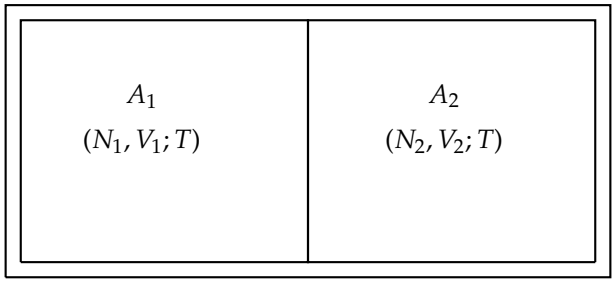


Figure 4.1: A schematic of two physical systems in thermal contact.

Gibbs visualized the mixing of two gases 1 and 2 with the same initial temperature T. Clearly, the temperature of the mixture would also be the same. For each gas, we compute the entropy as follows,

$$S_i = N_i k\ln V_i + \frac{2}{3}N_i k\left[1 + \ln(\frac{2\pi m_i kT}{h^2})\right] \tag{4.2}$$

After mixing, the total is,

$$S_T = \sum_{i=1}^{2}\left[N_i k\ln V + \frac{2}{3}N_i k\left[1 + \ln(\frac{2\pi m_i kT}{h^2})\right]\right] \tag{4.3}$$

where $V = V_1 + V_2$. Thus the net increase in the value of S called entropy of mixing is

$$\Delta S = S_T - \sum_{i=1}^{2} S_i = k\left[N_1 \ln \frac{V_1 + V_2}{V_1} + N_2 \ln \frac{V_1 + V_2}{V_2}\right] \tag{4.4}$$

The quantity ΔS is indeed positive according to 2nd thermodynamic law. For this case when two gases have same density, the equation becomes,

$$\Delta S^* = k\left[N_1 \ln \frac{N_1 + N_2}{N_1} + N_2 \ln \frac{N_1 + N_2}{N_2}\right] \tag{4.5}$$

Now, if we further assume that gas 1 and gas 2 has same N, V, T. By this equation, it seems that the system will gain some entropy after mixing. But we know this is not physical!

Gibbs realized that one can avoid this issue by simply adding one more term in the original expression of S. This is due to that we slip the multiplicity function to account for the fact that molecules in a gas are *indistinguishable*, and the factor is exactly $N!$.

$$\begin{aligned} S(N,V,E) &= Nk\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}Nk - k\ln N! \\ &= Nk\ln\left[\frac{V}{h^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{3}{2}Nk - k(N\ln N - N) \\ &= Nk\ln\left[\frac{V}{Nh^3}\left(\frac{4\pi mE}{3N}\right)^{3/2}\right] + \frac{5}{2}Nk \end{aligned} \tag{4.6}$$

Note here we used the Stirling approximation

$$\ln N! = N \ln N - N \tag{4.7}$$

In this case,

$$\Delta S^* = k\left[(N_1 + N_2)\ln\frac{V_1 + V_2}{N_1 + N_2} + N_1 \ln \frac{V_1}{N_1} + N_2 \ln \frac{V_2}{N_2}\right] \tag{4.8}$$

Therefore, the paradox could be resolved by counting. The equation is generally referred as the *Sackur − Tetrode* equation.

Let us see some consequences due to Gibbs's revision. First of all, we compute the energy,

$$E(S,V,N) = \frac{3h^2 N}{4\pi m V^{2/3}} \exp\left(\frac{2S}{3Nk} - 1\right) \qquad \text{previous} \tag{4.9}$$

$$E(S,V,N) = \frac{3h^2 N^{5/3}}{4\pi m V^{2/3}} \exp\left(\frac{2S}{3Nk} - \frac{5}{3}\right) \qquad \text{new} \tag{4.10}$$

This will make E as a truly extensive quantity. The most important consequence is on the chemical potential,

$$\mu = \left(\frac{\partial E}{\partial N}\right)_{V,S} = E\left[\frac{5}{3N} - \frac{2S}{3N^2 k}\right] \tag{4.11}$$

Applying the ideal gas law, we obtain

$$\mu = \frac{1}{N}[E + PV - TS] = \frac{G}{N} \tag{4.12}$$

Where G is the Gibbs free energy of the system. Express μ as a function of (N, V, T)

$$\mu(N, V, T) = kT\ln\left[\frac{N}{V}\left(\frac{h^2}{2\pi mkT}\right)^{3/2}\right] \tag{4.13}$$

We can also obtain F by $G - PV$

At room temperature and atmospheric pressure, $V/N = 4.2 \times 10^{26} m^3$, $(h^2/2\pi mkT)^{2/3}$ for helium is $1.3 \times 10^{-31} m^3$ Therefore, the logarithm is 12.7. Therefore, the chemical potential is -0.32 eV (helium) and -0.42 eV (for argon), respectively.

Chemical potential is usually negative!

4.2 How to count the microstates

In the previous section, we saw that the entropy of an N-particle system has to be divided by the factor of $k\ln(N!)$, which implies an *ad hoc* reduction of the number of microstates. It is natural to ask why? This is because that we were assuming that the particles are distinguishable. However, it is not true when we are considering the mixing of two same gases! To elaborate the point, we may say that if we consider two microstates that differ from another merely in an interchange of two particles in different energy states, then according to our original mode of counting, we would regard these microstates as distinct.

Suppose there exist n_1 particles being in the state of ϵ_1, n_2 particles being in the state of ϵ_2, ..., and $n1 + n2 + ... = N$. The total number of permutations that can be effected among N particles distributed according to the set $\{n_i\}$

$$\frac{N!}{n_1!n_2!...} \tag{4.14}$$

This is correct for situations in which all N particles happen to be in different energy states but is certainly wrong for other situations. Why?

The recipe of Gibbs corrects the enumeration of the microstates only in a gross manner. Numerically, this would approach closer and closer to reality as the probability of n_i being greater. This in turn happens when the system at high temperature (so that many energy states are accessible) and has low density. However, quantum statistics will require better treatment.

Physics 468/668: Statistical Physics **Spring 2019**

Lecture 5: Phase space and Liouville's theorem

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

5.1 Elements of Ensemble Theory

For a given macrostate (N, V, E), a statistical system at any time, is equally likely to be in any one of an very large number of distinct microstates. When the system continually switches from one microstate to another over a span of time, we actually observe a behavior averaged over the possible microstates. Then, we expect the averaged behavior of any system in this collection as an *ensemble*.

5.2 Phase space

For a given classical system, at any time t, it may be defined by specifying the instantaneous positions and momenta of all the particles constituting the system. Let the number of particles as N, the definition of a microstate requires the information of $3N$ positions $(q_1, q_2, ..., q_{3N})$ and $3N$ momentum coordinates $(p_1, p_2, ..., p_{3N})$. Hence, we define the set of (q_i, p_i), where $i = 1, 2, ..., 3N$, as a point in a space of $6N$ dimensions. We refer to this space as the *phase space*, and the phase point (q_i, p_i) as a representative points of the given system.

$$\begin{aligned} \dot{q_i} &= \frac{\partial H(q_i, p_i)}{\partial p_i} \\ \dot{p_i} &= -\frac{\partial H(q_i, p_i)}{\partial q_i} \end{aligned} \tag{5.1}$$

where $H(q_i, p_i)$ is the *Hamiltonian* of the system. With the time passes, the set of (q_i, p_i) undergoes a continual change. Correspondingly, it forms a trajectory depending on the velocity v vector $\equiv (\dot{q_i}, \dot{p_i})$, which in turn is given by the above equation. It is not difficult to see that the trajectory of the points must remain within a limited region of the phase space, because a finite volume V directly limits the values of the coordinates q_i, and E limits the values of both q_i and p_i. In particular, the total energy of the system is known to satisfy the following,

$$H(q_i, p_i) = E. \tag{5.2}$$

This means that the trajectory needs to be restricted to the hyper-surface defined by E. If we allow the energy to vary, say $E \pm \Delta$, the corresponding trajectory will be restricted to a hyper-shell defined by the limits.

To describe the overall movements of points as a function of time, we can define a *probability density function* $\rho(p_i, q_i, t)$. This function is such that at any time t, the number of points in the volume element $(d^{3N}q d^{3N}p)$ around (q, p) is given by

$$\int \rho(p_i, q_i, t) d^{3N}q d^{3N}p \tag{5.3}$$

Similarly, any physical quantity could be described as the averaged values along the phase space.

$$< f >= \frac{\int f(q,p)\rho(q,p)d^{3N}qd^{3N}p}{\int \rho(p_i,q_i,t)d^{3N}qd^{3N}p} \tag{5.4}$$

Note that the ensemble average $< f >$ itself is also a function of time.

An ensemble is said to be stationary if ρ does not depend explicitly on time, namely,

$$\frac{\partial \rho}{\partial t} = 0 \tag{5.5}$$

Clearly, for such an ensemble, the $< f >$ will be independent of time as well. Naturally, this represents an *equilibrium* state.

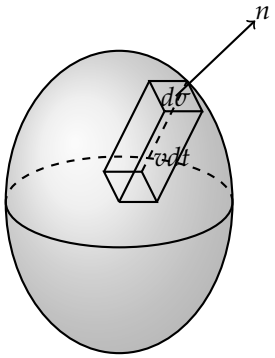


Figure 5.1: A schematic of phase space.

5.3 Liouville's theorem

Consider an arbitrary volume ω in the phase space and let the surface enclosing this volume be denoted by σ. Then the number of points in this volume increases with time,

$$\frac{\partial}{\partial t}\int_\omega \rho(t)d\omega \tag{5.6}$$

where $d\omega \equiv (d^{3N}qd^{3N}p)$.

On the other hand, the net rate at which the points flow out of ω is given by

$$\int_\sigma \rho \mathbf{v}\cdot\vec{n}d\sigma \tag{5.7}$$

Here, $\mathbf{v}$ is the velocity vector of the representative points in the region of the surface element $d\sigma$ while $\vec{n}$ is the outward unit vector normal to this element. By the divergence theorem,

$$\int_\sigma \mathrm{div}(\rho\mathbf{v})d\omega \tag{5.8}$$

$$\mathrm{div}(\rho\mathbf{v}) \equiv \sum_{i=1}^{3N}\left[\frac{\partial}{\partial q_i}(\rho\dot{q}_i) + \frac{\partial}{\partial p_i}(\rho\dot{p}_i)\right] \tag{5.9}$$

The number of points should be conserved, we have

$$\frac{\partial\rho}{\partial t}\int_\omega d\omega \equiv -\int_\omega \mathrm{div}(\rho\mathbf{v})d\omega \tag{5.10}$$

that is

$$\int_\omega\left[\frac{\partial\rho}{\partial t} + \mathrm{div}(\rho\mathbf{v})\right]d\omega = 0. \tag{5.11}$$

The necessary and sufficient condition to let the integral vanish is that the integrand itself vanish everywhere,

$$\frac{\partial\rho}{\partial t} + \mathrm{div}(\rho\mathbf{v}) = 0 \tag{5.12}$$

Let's break down the divergence term,

$$\frac{\partial\rho}{\partial t} + \sum_i^{3N}(\frac{\partial\rho}{\partial q_i}\dot{q}_i + \frac{\partial\rho}{\partial p_i}\dot{p}_i) + \rho\sum_i^{3N}(\frac{\partial\dot{q}_i}{\partial q_i} + \frac{\partial\dot{p}_i}{\partial p_i}) = 0 \tag{5.13}$$

The last group of terms vanishes identically because ,

$$\frac{\partial\dot{q}_i}{\partial q_i} = \frac{\partial^2 H(q_i,p_i)}{\partial q_i\partial p_i} \equiv \frac{\partial^2 H(q_i,p_i)}{\partial p_i\partial q_i} = -\frac{\partial\dot{p}_i}{\partial p_i} \tag{5.14}$$

Further, since $\rho \equiv \rho(q,p,t)$, the remaining term may be combined to form the total derivative of ρ as a function of t, with the results that,

$$\frac{d\rho}{dt} = \frac{\partial\rho}{\partial t} + [\rho, H] = 0 \tag{5.15}$$

where we use the *Poission braket* $[\rho, H]$,

$$[\rho, H] = \sum_{i=1}^{3N}\left(\frac{\partial\rho}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial\rho}{\partial p_i}\frac{\partial H}{\partial q_i}\right) \tag{5.16}$$

According to the theorem, the total number of systems in the ensemble is a constant. Thus the swarm of the representing points moves in the phase space in essentially the same manner as an incompressible fluids moves in the physical space!

If the density is independent of the coordinates (q,p) over the region of phase space, physically, this choice corresponds to an ensemble of systems that at all times are uniformly distributed over all possible microstates. The resulting ensemble is *microcanonical* ensemble, in which the averaged properties could be determined by the following,

$$< f >= \frac{\int f(q,p)\rho(q,p)d^{3N}qd^{3N}p}{\int\rho(p_i,q_i,t)d^{3N}qd^{3N}p} = \frac{1}{\omega}\int_\omega f(q,p)d\omega \tag{5.17}$$

A more general way of satisfying the condition is to assume that the dependence of ρ on (q, p) comes only through an explicit dependence on H,

$$\rho(q, p) = \rho[H(q, p)] \tag{5.18}$$

This equation provides a class of density functions for which the corresponding ensemble is stationary. The most natural choice of such ensemble is the one for which

$$\rho(q, p) \propto \exp[-H(q, p)/kT] \tag{5.19}$$

which is *canonical* ensemble.

5.4 Homework

1. prove that

$$\mu(N, V, T) = kT\ln\left[\frac{N}{V}\left(\frac{h^2}{2\pi mkT}\right)^{3/2}\right] \tag{5.20}$$

   and calculate μ for the following gases (He, Ne, Ar, Kr, Xe) at room temperature and ambient pressure conditions.

2. 1.12
3. 1.13
4. 1.14
5. 1.15

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Lecture 6: The Microcanonical ensemble

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

6.1 Properties

In this ensemble, the macrostate of a system is defined by (N, V, E). However, in view of the considerations in the previous lecture, we may prefer to specify a range of energy values, say from $(E - 1/2\Delta)$ to $(E + 1/2\Delta)$. With the macrostate specified, a choice still remains for a system to be in any one of a large number of possible microstates. In the phase space, the representative points of the ensemble have a choice to lie anywhere within a 'hypershell' defined by the conditions.

$$E - \frac{1}{2}\Delta \leq H(q,p) \leq E + \frac{1}{2}\Delta \tag{6.1}$$

The volume of the phase space enclosed within this shell is given by

$$\omega = \int d\omega = \int (d^{3N}q d^{3N}p) \tag{6.2}$$

It is clear that ω would be a function of (N, V, E, Δ).

Now, we consider the density function,

$$\begin{aligned} \rho(q,p) &= \text{const} \quad \text{if} \quad (E - 1/2\Delta) \leq H(q,p) \leq (E + 1/2\Delta) \\ &= 0 \qquad \text{otherwise} \end{aligned} \tag{6.3}$$

Accordingly, we expect that the number of representative points lying in a hypershell with a thickness of Δ.

Since the ensemble under study is a stationary one, the ensemble average of any property f will be independent of time. The next thing is to establish a connection between the mechanics and thermodynamics of the systems. The volume ω is a direct measurement of the multiplicity (Γ) of the microstates accessible to the system.

Let's say,

$$\Gamma = \omega/\omega_0 \tag{6.4}$$

The thermodynamics of the system would then follow in the same way as

$$S = k\ln\Gamma = k\ln(\omega/\omega_0) \tag{6.5}$$

The problem then is to determine ω_0. From the dimensional considerations, ω_0 must be in the nature of an *angular momentum raised to the power 3N*. This could be determined exactly for a few simple systems.

6.2 Ideal gas

For ideal gas in the micro-canonical ensemble, the volume ω of the phase space is given by

$$\omega = \int \cdots \int (d^{3N}q d^{3N}p) \tag{6.6}$$

Since the Hamiltonian in this case is a function of the p_i alone, integrations over q_i could be carried out as follows,

$$\omega = \int \cdots \int_{E-\frac{1}{2}\Delta \le \sum_{i=1}^{3N}(p_i^2/2m) \le E+\frac{1}{2}\Delta} d^{3N}p = \int \cdots \int_{2m(E-\frac{1}{2})\Delta \le \sum_{i=1}^{3N}(p_i^2) \le 2m(E+\frac{1}{2}\Delta)} d^{3N}y \tag{6.7}$$

This is equal to the volume of $3N$-dimensional hyper-shell bounded by a hypersphere of radii

$$\sqrt{2m(E-\frac{1}{2})\Delta} \le r \le \sqrt{2m(E+\frac{1}{2}\Delta)} \tag{6.8}$$

If $\Delta \ll E$, this is given by the thickness of the shell, which is almost equal to $\Delta(m/2E)^{1/2}$, multiplied by the surface area of a $3N$-dimensional hyper-shphere. By the equation of (7) of Appendix C, we obtain the integral,

$$\Delta\left(\frac{m}{2E}\right)^{1/2}\left[\frac{2\pi^{3N/2}}{[(3N/2-1)]!}(2mE)^{(3N-1)/2}\right] \tag{6.9}$$

Therefore,

$$\omega \approx \frac{\Delta}{E}V^N \frac{(2\pi mE)^{3N/2}}{[(3N/2-1)!]} \tag{6.10}$$

Comparing this with the following

$$\omega_0 = h^{3N} \tag{6.11}$$

Where the upper-script denotes the number of degrees of freedom.

In the case of a single particle (N=3), the number of microstates available would asymptotically be equal to the volume of allowed region divided by h^3. We then calculate the number of microstates available to a free particle confined to volume V and its momentum p being less than or equal to a given value P,

$$\sum(P) \approx \frac{1}{h^3}\int \cdots \int_{p \le P}(d^3q d^3p) = \frac{V}{h^3}\frac{4\pi}{3}P^3, \tag{6.12}$$

from which we obtain for the number of microstates with momentum lying between p and $p+dp$,

$$g(p)dp = \frac{d\sum(p)}{dp}dp \approx \frac{V}{h^3}4\pi p^2 dp \tag{6.13}$$

Expressed in terms of the particle energy, these expressions assume the form

$$\sum(E) \approx \frac{V}{h^3}\frac{4\pi}{3}(2mE)^{3/2} \tag{6.14}$$

and

$$g(E)dE = \frac{d\sum(E)}{dE}dE \approx \frac{V}{h^3}2\pi(2m)^{3/2}E^{1/2}dE \tag{6.15}$$

6.3 1D simple harmonic oscillator

For this system, we have the Hamiltonian as follows

$$H(q,p) = \frac{1}{2}kq^2 + \frac{1}{2m}p^2 \tag{6.16}$$

where k is the spring constants. The space coordinate q and the momentum coordinate p of the system are given by

$$q = A\cos(\omega t + \phi) \tag{6.17}$$

$$p = m\dot{q} = -m\omega A\sin(\omega t + \phi) \tag{6.18}$$

Where A is the amplitude and ω is the angular frequency of vibration;

$$\omega = \sqrt{k/m} \tag{6.19}$$

And the energy of the oscillator is a constant of the motion,

$$E = \frac{1}{2}m\omega^2 A^2 \tag{6.20}$$

The phase-space trajectory of the representative point (q,p) of the system is determined by eliminating t ,

$$\frac{q^2}{(2E/m\omega^2)} + \frac{p^2}{2mE} = 1 \tag{6.21}$$

which is an ellipse, with axes proportional to $\sqrt{E}$. Now, if we restrict the oscillator energy to the interval $[E - 1/2\Delta, E + 1/2\Delta]$, the representative points in the phase space will be confined to the region bounded by elliptical trajectories corresponding to the energy values $E - 1/2\Delta$ and $E + 1/2\Delta$. The volume will be

$$\int\cdots\int_{E-\frac{1}{2}\Delta\le H(q,p)\le E+\frac{1}{2}\Delta} dqdp = \frac{2\pi(E+\frac{1}{2}\Delta)}{\omega} - \frac{2\pi(E-\frac{1}{2}\Delta)}{\omega} = \frac{2\pi\Delta}{\omega} \tag{6.22}$$

According to quantum mechanics, the energy eigenvalues are,

$$E_n = (n + \frac{1}{2})\bar{h}\omega \tag{6.23}$$

In terms of phase phase, one could say that the representative point of the system must move along one of the 'chosen' trajectories; the area of the phase space between two consecutive trajectories, for which $\Delta = \bar{h}w$ is simply $2\pi\hbar$. For $E \gg \Delta \gg \hbar\omega$, the number of eigenstates within the allowed energy interval is nearly equal to $\Delta/\hbar\omega$. Hence, the area of the phase space equivalent to one eigenstate is given by,

$$\omega_0 = (2\pi\Delta/\omega)/(\Delta/\bar{h}w) = 2\pi\bar{h} = h \tag{6.24}$$

If we consider a system of N harmonic oscillators along the same lines, we arrive at the results: $\omega_0 = h^N$.

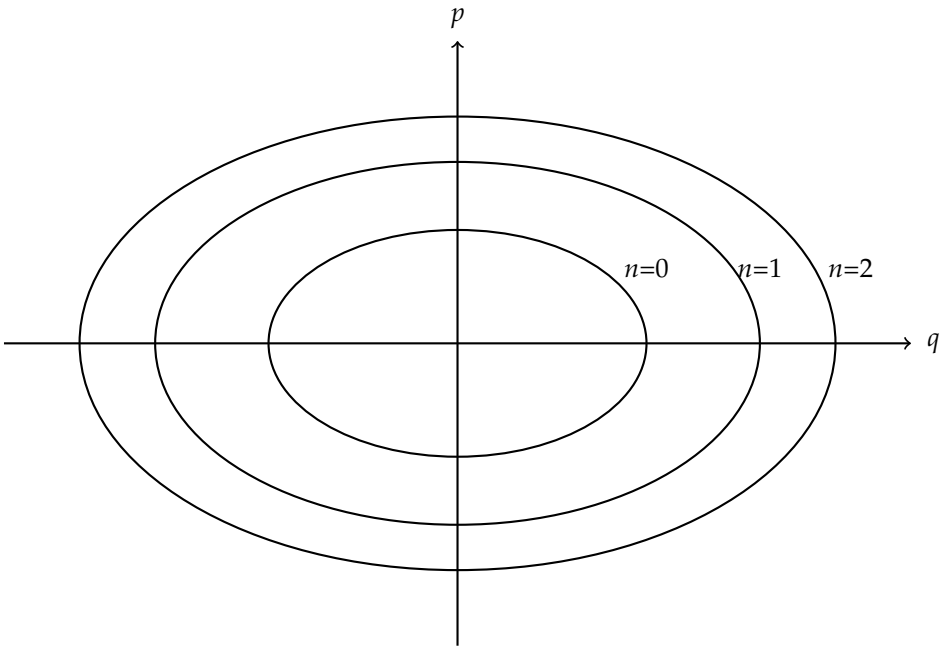


Figure 6.1: A schematic of phase space.

6.4 Homework

1. 2.3
2. 2.5
3. 2.7
4. 2.8
5. 2.9

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Lecture 7: The Canonical Ensemble

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

7.1 Background

We established the basis of ensemble theory and made a detailed study of the micro-canonical ensembles in chapter 2. In that ensemble, the macrostate of the systems was defined through a fixed (N, V, E), where we allow E has a fluctuation of Δ. The basic problem is to determine $\Omega(N, V, E, \Delta)$. For some systems, the complete thermodynamics can be derived.

The concept of a fixed energy (even with a range) is not very practical. First, the total energy E of a system is hardly to be measured. Second, it is hard to keep its value under strict physical control. In this sense, it is more practical to speak of a fixed temperature. Such an ensembel is referred to as a canonical ensemble.

In canonical ensemble, the energy E of a system variable,. It can take any value between 0 and infinity. The consequent question is, *what is the probability P_r that, at any time, a system is found to be one of the state characterized by the energy value E_r?*

There are two ways to determine the relation.

1. one consists of regarding the system as in equilibrium with a heat reservior at a common T
2. one consists of regarding the system as in a member of a canonical ensemble (N, V, T) in contact with many others.

We expect that in the thermodynamic limit the final results would be the same in both cases. Once P_r is determined, the rest follows without difficulty.

7.2 Equilibrium between a system and a heat reservoir

In a heat reservoir, the system and the reservoir would have a common T. Their energies, however, would be variable, in principle could have, at any time, values between 0 and $E^{(0)}$, where $E^{(0)}$ denotes the energy of the composite system $A + A'$. If the system A happens to be in a state characterized by the energy value E_r, the reservoir would have an energy E'_r, such that

$$E_r + E'_r = E^{(0)} = \text{const} \tag{7.1}$$

We assume that the reservoir is much larger than the given system, thus we have

$$\frac{E_r}{E^{(0)}} = \left(1 - \frac{E'_r}{E^{(0)}}\right) \ll 1 \tag{7.2}$$

When the state of A (E_r) is specified, the reservoir A′ can still be in any one of a large number of states compatible with the energy value E'_r. Let the number of these states be denoted by $\Omega'(E'_r)$. The prime on the symbols emphasizes the fact that its functional form will depend on the reservoir. Since the various possible states are equally likely to occur, the relevant probability would be directly proportional to this number.

$$P_r \propto \Omega'(E'_r) = \Omega'(E^{(0)} - E_r) \tag{7.3}$$

Expanding the logarithm of $\Omega'(E'_r)$,

$$\ln \Omega'(E'_r) = \ln \Omega'(E^{(0)}) + \left(\frac{\partial \ln \Omega'}{\partial E'}\right)_{E'=E^{(0)}} (E'_r - E^{(0)}) + ... \approx \text{const} - \beta' E_r \tag{7.4}$$

where we defined β in the previous lecture,

$$\beta = \left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V} \tag{7.5}$$

Note that, in equilibrium, $\beta = \beta' = 1/kT$. From the above equations, we obtain the desired results,

$$P_r \propto \exp(-\beta E_r) \tag{7.6}$$

After normalization, we get

$$P_r = \frac{\exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} \tag{7.7}$$

7.3 A system in the canonical ensemble

We consider an ensemble of N identical systems (0,1,2,...N) sharing a total energy E; let E_r denote the energy eigenvalues of the systems. If n_r denotes the number of systems which, at any time t, have the energy value E_r, then the set of numbers n_r must satisfy the conditions,

1. n_1 systems have energy of E_1
2. n_2 systems have energy of E_2
3. n_3 systems have energy of E_3
4. .....

$$\sum_r n_r = N \tag{7.8}$$

$$\sum_r n_r E_r = E = NU \tag{7.9}$$

where U denotes the average energy per system in the ensemble. Any set n_r that satisfies the above conditions is a possible state. Such state can be realized in a number of ways by permutation, therefore, the total number is

$$W\{n_r\} = \frac{N!}{n_0! n_1! \cdots} \tag{7.10}$$

All possible states are equally likely to occur, the frequency with a given $\{n_r\}$ will be directly proportional to the number $W\{n_r\}$. With a very large number of particles, we will find the distribution of $W\{n_r\}$ is very localized. That is, the most probable distribution of $\{n_r\}$ which lead to the maximum W is the only possible state.

7.4 The most probable values

In order to maximize W, we choose to work with $\ln W$,

$$\ln W\{n_r\} = \ln N! - \sum_r \ln n_r! \tag{7.11}$$

To the limit $N \to \infty$, the values of n_r would also tend to infinity. Applying the Stirling approximation,

$$\ln W\{n_r\} = N\ln N - \sum_r n_r \ln n_r \tag{7.12}$$

If we shift from the set $\{n_r\}$ to a slightly different set $\{n_r + \delta n_r\}$, the expression would change to

$$\delta(\ln W\{n_r\}) = -\sum_r \delta n_r(\ln n_r + 1) \tag{7.13}$$

If the set $\{n_r\}$ is maximal, the variation $\delta(\ln W\{n_r\})$ should vanish. At the same time, it also satisfies,

$$\sum_r \delta n_r = 0 \tag{7.14}$$

$$\sum_r \delta n_r E_r = 0 \tag{7.15}$$

Therefore, the desired set $\{n_r\}^*$ is determined by the method of Lagrange multipliers, by which the condition becomes,

$$\sum_r (-(\ln n_r^* + 1) - \alpha - \beta E_r)\delta n_r = 0 \tag{7.16}$$

Where α and β are coefficients due to Lagrange multipliers. The only way to satisfy the condition is to let all coefficients vanish identically,

$$\ln n_r^* = -(\alpha + 1) - \beta E_r \tag{7.17}$$

which gives

$$n_r^* = C\exp(-\beta E_r) \tag{7.18}$$

To determine C and β, we subject the equation to the original condition,

$$\frac{n_r^*}{N} = \frac{\exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} \tag{7.19}$$

The parameter β should be a solution of the equation,

$$U = \frac{\sum_r \exp E_r(-\beta E_r)}{\sum_r \exp(-\beta E_r)} \tag{7.20}$$

7.5 The mean values

To evaluate this expression, we replace the weight factors

$$\tilde{W}\{n_r\} = \frac{N! w_0^{n_0} w_1^{n_1} w_2^{n_2} \cdots}{n_0! n_1! n_2! \cdots} \tag{7.21}$$

$$\Gamma(N,U) = \sum{}' \tilde{W}\{n_r\} \tag{7.22}$$

TODO

7.6 Statistic quantities in the canonical ensemble

With the canonical distribution, we now look for a general recipe to extract information about the various macroscopic properties of the given system. Remember the definition of Helmholtz free energy $A = U - TS$

$$dA = dU - TdS - SdT = -SdT - PdV + \mu dN \tag{7.23}$$

$$S = -\left(\frac{\partial A}{\partial T}\right)_{N,V} \tag{7.24}$$

$$P = -\left(\frac{\partial A}{\partial V}\right)_{N,T} \tag{7.25}$$

$$\mu = -\left(\frac{\partial A}{\partial N}\right)_{V,T} \tag{7.26}$$

and

$$U = A + TS = A - T\left(\frac{\partial A}{\partial T}\right)_{N,V} = -T^2\left[\frac{\partial}{\partial T}\left(\frac{A}{T}\right)\right]_{N,V} = \left[\frac{\partial (A/T)}{\partial (1/T)}\right]_{N,V} \tag{7.27}$$

Comparing this equation with

$$U = \frac{\sum_r E_r \exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} = -\frac{\partial}{\partial \beta} \ln\left[\sum_r \exp(-\beta E_r)\right] \tag{7.28}$$

we find there is a close relation that

$$-\frac{A}{kT} = \ln\left[\sum_r \exp(-\beta E_r)\right] \tag{7.29}$$

The equation gives that

$$A(N,V,T) = -kT \ln Q_N(V,T) \tag{7.30}$$

where $Q_N(V,T)$ is the partition function of the system.

$$Q_N(V,T) = \sum_r \exp(-E_r/kT) \tag{7.31}$$

Once A is known, one could derive the rest quantities such as S, P, μ, and C_V as well,

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{N,V} = -T\left(\frac{\partial^2 A}{\partial T^2}\right)_{N,V} \tag{7.32}$$

$$G = A + PV = A - V\left(\frac{\partial A}{\partial V}\right)_{V,T} = N\left(\frac{\partial A}{\partial N}\right)_{V,T} = N\mu \tag{7.33}$$

The entropy is the system is determined by $P_r = Q^{-1}\exp(-\beta E_r)$

$$< \ln P_r >= -\ln Q - \beta < E_r >= \beta(A - U) = -S/k \tag{7.34}$$

Thus

$$S = -k < \ln P_r >= -k\sum_r P_r \ln P_r \tag{7.35}$$

This shows that the entropy is solely and completely determined by the probability P_r! When T is 0 K, P_r is equal to 1 for this state and 0 for the rest. Then the entropy is exactly 0. We also infer that the vanishing entropy and perfect statistical order go together. When the number of accessible states rises, the entropy of the system increases because more P_r appears.

For micro-canonical ensemble, $P_r = 1/\Omega$ Thus,

$$S = -k\sum_{r=1}^{\Omega}\frac{1}{\Omega}\ln(\frac{1}{\Omega}) = k\ln\Omega \tag{7.36}$$

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Lecture 8: The Canonical Ensemble

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

8.1 Density of States

In most physical cases the energy levels accessible to a system is degenerate, that is, one has a group of states g_i, belonging to the same energy value E_i. In such cases, it is more convenient to write the partition function as

$$Q_N(V,T) = \sum_i g_i \exp(-\beta E_i) \tag{8.1}$$

The corresponding expression for P_i would be

$$P_i = \frac{g_i \exp(-\beta E_i)}{\sum_i g_i \exp(-\beta E_i)} \tag{8.2}$$

Clearly, the g_i states are all equally likely to occur. As a result, the probability of a system having energy E_i becomes proportional to the multiplicity g_i of this level.

In view of the largeness of the number of particles constituting a given system and the largeness of the volume to which these particles are confined, the consecutive energy values E_i of the system are, in general, very close to one another. Accordingly, there lie, within any reasonable interval of energy $(E, E+dE)$, a very large number of energy levels.

Denoting it by $g(E)dE$, where $g(E)$ denotes the density of states around energy value E, we have

$$P(E)dE \propto \exp(-\beta E) g(E) dE \tag{8.3}$$

After normalization,

$$P(E)dE = \frac{\exp(-\beta E) g(E) dE}{\int_0^\infty \exp(-\beta E) g(E) dE} \tag{8.4}$$

The denominator here is yet another expression for the partition function,

$$Q_N(V,T) = \int_0^\infty \exp(-\beta E) g(E) dE \tag{8.5}$$

With $\beta > 0$, the partition function $Q(\beta)$ is just the Laplace transform of $g(E)$. We may therefore, write $g(E)$ as the inverse Laplace transform of $Q(\beta)$

$$\begin{aligned} g(E) &= \frac{1}{2\pi i} \int_{\beta'-i\infty}^{\beta'+i\infty} \exp(\beta E) Q(\beta) d\beta \ (\beta' > 0) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp[(\beta' + i\beta'')E] Q(\beta' + i\beta'') d\beta'' \end{aligned} \tag{8.6}$$

Accordingly, the averaged value could be written as,

$$< f >= \sum_i f_i P_i = \frac{\sum_i f(E_i) g_i \exp(-\beta E_i)}{\sum_i g_i \exp(-\beta E_i)} \rightarrow \frac{\int_0^\infty f(E) \exp(-\beta E) g(E) dE}{\int_0^\infty \exp(-\beta E) g(E) dE} \tag{8.7}$$

8.2 The classical systems

The theory developed in the preceding sections is of general applicability. We recall the concepts in ensemble theory, the averaged quantity $< f >$ is,

$$< f >= \frac{\int f(q,p)\rho(q,p) d^{3N}q d^{3N}p}{\int \rho(q,p) d^{3N}q d^{3N}p} \tag{8.8}$$

For canonical system, it follows,

$$\rho(q,p) \propto \exp[-\beta H(q,p)] \tag{8.9}$$

The expression for $< f >$ then take the form

$$< f >= \frac{\int f(q,p) \exp(\beta H) d\omega}{\int \exp(\beta H) d\omega} \tag{8.10}$$

The denominator of this expression is directly related to the partition function of the system. However, we must take into account the relationship between a volume element in the phase space, and the corresponding number of distinct quantum states of the system. The relationship is

$$\frac{d\omega}{N! h^{3N}} \tag{8.11}$$

The appropriate expression for the partition function would be,

$$Q_N(V,T) = \frac{1}{N! h^{3N}} \int \exp(-\beta H(q,p)) d\omega \tag{8.12}$$

We consider an ideal gas with N identical molecules, confined to V and in equilibrium at T. Since there are no intermolecular interactions to be taken into account, the energy is

$$H(q,p) = \sum_{i=1}^{N} (p_i^2/2m) \tag{8.13}$$

The partition function of the system would be then,

$$Q_N(V,T) = \frac{1}{N! h^{3N}} \int \exp\left[-(\beta/2m) \sum_i p_i^2 \right] \prod_{i=1}^{N} (d^3 q_i d^3 p_i) \tag{8.14}$$

Integration over the space gives V^N. Integrations over the momentum space are also trivial,

$$Q_N(V,T) = \frac{V^N}{N! h^{3N}} \left[\int_0^\infty \exp(-p^2/2mkT)(4\pi p^2 dp) \right]^N = \frac{1}{N!} \left[\frac{V}{h^3} (2\pi mkT)^{3/2} \right]^N \tag{8.15}$$

The Helmholtz energy is then given by

$$A(N,V,T) = -kT \ln Q_N(V,T) = NkT\left[\ln\left(\frac{N}{V}(\frac{h^2}{2\pi mkT})^{3/2}\right) - 1\right] \tag{8.16}$$

With knowing A, we can get the rest properties,

$$\mu = \left(\frac{\partial A}{\partial N}\right)_{V,T} = kT \ln\left[\frac{N}{V}\left(\frac{h^2}{2\pi mkT}\right)^{3/2}\right] \tag{8.17}$$

$$P = \left(\frac{\partial A}{\partial V}\right)_{N,T} = \frac{NkT}{V} \tag{8.18}$$

$$S = \left(\frac{\partial A}{\partial T}\right)_{N,V} = Nk\left[\ln\left(\frac{N}{V}(\frac{h^2}{2\pi mkT})^{3/2}\right) + \frac{5}{2}\right] \tag{8.19}$$

The results are similar to what we derived from thermodynamics.

At this stage, we may write

$$Q_N(V,T) = \frac{1}{N!}[Q_1(V,T)]^N \tag{8.20}$$

Going back to the ideal gas, we could as well start with the density of states $g(E)$.

$$g(E) = \frac{\partial \sum}{\partial E} \approx \frac{1}{N!}\left(\frac{V}{h^3}\right)^N \frac{(2\pi m)^{3N/2}}{[(3N/2-1)]!} E^{3N/2-1} \tag{8.21}$$

Substituting this into eq(*), we obtain

$$Q_N(\beta) = \frac{1}{N!}\left(\frac{V}{h^3}\right)^N \left(\frac{2\pi m}{\beta}\right)^{3N/2} \tag{8.22}$$

If we start with the single particle, we will get

$$\alpha(\epsilon) = \frac{2\pi V}{h^3}(2m)^{3/2}\epsilon^{1/2} \tag{8.23}$$

And the corresponding single-particle partition function

$$Q_1(\beta) = \int_0^\infty \exp(-\beta\epsilon)\alpha(\epsilon)d\epsilon = \frac{V}{h^3}\left(\frac{2\pi m}{\beta}\right)^{3/2} \tag{8.24}$$

How to calculate $g(E)$ from Q? Assuming that Q is known, we have

$$g(E) = \frac{V^N}{N!}\left(\frac{2\pi m}{h^2}\right)^{3N/2} \frac{1}{2\pi i}\int_{\beta'+i\infty}^{\beta'-i\infty} \frac{\exp(\beta E)}{\beta^{3N/2}} d\beta \tag{8.25}$$

Noting that for all positive n,

$$\frac{1}{2\pi i}\int_{s'-i\infty}^{s'+i\infty} \frac{\exp(sx)}{s^{n+1}} ds = \begin{cases} \frac{x^n}{n!} & \text{if } x \geq 0 \\ 0 & \text{if } x \leq 0 \end{cases} \tag{8.26}$$

Therefore

$$g(E) = \begin{cases} \frac{V^N}{N!}\left(\frac{2\pi m}{h^2}\right)^{3N/2} \frac{E^{(3N/2)-1}}{(3N/2-1)!} & \text{if } E \geq 0 \\ 0 & \text{if } E < 0 \end{cases} \tag{8.27}$$

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Lecture 9: Energy fluctuation and equipartition theorem

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

9.1 Energy fluctuation

We said that in the canonical ensemble, a system could take energy anywhere between zero and infinity. However, we don't know how it works in reality.

To answer this question, let's examine the range based on the expression of the mean energy.

$$U =< E >= \frac{\sum_r E_r \exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} \tag{9.1}$$

Differentiating it with respect to β, we get,

$$\frac{\partial U}{\partial \beta} = -\frac{\sum_r E_r^2 \exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} + \frac{\left[\sum_r E_r \exp(-\beta E_r)\right]^2}{\left[\sum_r \exp(-\beta E_r)\right]^2} = - < E^2 > + < E >^2 \tag{9.2}$$

from which it follows,

$$< E^2 > - < E >^2 = -\frac{\partial U}{\partial \beta} = kT^2\left(\frac{\partial U}{\partial T}\right) = kT^2 C_V \tag{9.3}$$

Therefore, the deviation is

$$\frac{\sqrt{< E^2 > - < E >^2}}{< E >} = \frac{\sqrt{kT^2 C_V}}{U} \tag{9.4}$$

Which is $O(N^{-1/2})$. Therefore, the fluctuation would become negligible when N is large.

If we consider the manner in which energy is distributed among the various members of the canonical ensemble. We treat E as a continuous variable,

$$P(E)dE \approx \exp(-\beta E) g(E) dE \tag{9.5}$$

The probability function is governed by two terms

1. the Boltzmann factor, which monotonically decreases with E
2. the density of states, which monotonically increases with E

Thereby, it must reach some extreme E^*

$$\left(\frac{\partial(\exp(-\beta E)g(E))}{\partial E}\right)|_{E=E^*} = 0 \tag{9.6}$$

That is

$$\frac{\partial \ln g(E)}{\partial E}|_{E=E^*} = \beta \tag{9.7}$$

Given that

$$S = k \ln g \tag{9.8}$$

and

$$\left(\frac{\partial S(E)}{\partial E}\right)_{E=U} = \frac{1}{T} = k\beta \tag{9.9}$$

$$E^* = U \tag{9.10}$$

This implies that the most likelihood value of energy is identical to the mean value, regardless of the physical nature of the given system.

We now expand the logarithm of $P(E)$ around E^*

$$\begin{aligned} \ln[\exp(-\beta E)g(E)] &= \left(-\beta U + \frac{S}{k}\right) + \frac{1}{2}\frac{\partial^2}{\partial E^2}\ln\left[\exp(-\beta E)g(E)\right]|_{E=U}(E-U)^2 + ... \\ &= -\beta(U - TS) - \frac{1}{2kT^2C_V}(E-U)^2 + ... \end{aligned} \tag{9.11}$$

From which we obtain,

$$P(E) \propto \exp(-\beta E)g(E) \approx \exp(-\beta(U-TS))\exp\left[-\frac{(E-U)^2}{2kT^2C_V}\right] \tag{9.12}$$

This is a Gaussian distribution, with mean value U and dispersion $\sqrt{kT^2C_V}$.

Take the example of ideal gas,

9.2 Equipartition Theorem

To derive these theorems, we determine the expectation value of the quantity $x_i(\partial H/\partial x_j)$, where $H(q,p)$ is the Hamiltonian of the system while x_i and x_j are any two of the $6N$ coordinates (q,p).

In the canonical ensemble,

$$\left\langle x_i \frac{\partial H}{\partial x_j}\right\rangle = \frac{\int (x_i \frac{\partial H}{\partial x_j})\exp(-\beta H)d\omega}{\int \exp(-\beta H)d\omega} \tag{9.13}$$

Integrating over x_j by parts, the numerator becomes,

$$\int\left[-\frac{1}{\beta}x_i e^{-\beta H}|_{(x_j)_1}^{(x_j)_2} + \frac{1}{\beta}\int\left(\frac{\partial x_i}{\partial x_j}e^{-\beta H}dx_j\right)\right]d\omega(j) \tag{9.14}$$

Here $(x_j)_1$ and $(x_j)_2$ are the extreme values of the coordinate x_j. The integrated part here vanishes because whenever any of the coordinates takes an extreme value the Hamiltonian of the system becomes infinite. In the integral that remains, the factor $\partial x_i/\partial x_j$, being equal to δ_{ij}, comes out of the integral sign and we are left with

$$\frac{1}{\beta}\delta_{ij}\int e^{-\beta H}d\omega \tag{9.15}$$

Substituting this into (1), we get,

$$\left\langle x_i\frac{\partial H}{\partial x_j}\right\rangle = \delta_{ij}kT \tag{9.16}$$

Which is independent of the precise form of the function H.

In the special case $x_j = x_i = p_i$, the above equation becomes,

$$\left\langle p_i\frac{\partial H}{\partial p_i}\right\rangle =< p_i\dot{q}_i >= kT \tag{9.17}$$

In the special case $x_j = x_i = q_i$, the above equation becomes,

$$\left\langle q_i\frac{\partial H}{\partial q_i}\right\rangle = - < q_i\dot{p}_i >= kT \tag{9.18}$$

Adding over all i, we obtain

$$\left\langle \sum_i p_i\frac{\partial H}{\partial p_i}\right\rangle =< \sum_i p_i\dot{q}_i >= 3NkT \tag{9.19}$$

$$\left\langle \sum_i q_i\frac{\partial H}{\partial q_i}\right\rangle = - < \sum_i q_i\dot{p}_i >= 3NkT \tag{9.20}$$

Now, if H is composed of quadratic functions,

$$H = \sum_j A_jP_j^2 + \sum_j B_jQ_j^2 \tag{9.21}$$

where P_j and Q_j are canonically conjugate, We clearly have

$$\sum_j(P_j\frac{\partial H}{\partial P_j} + Q_j\frac{\partial H}{\partial Q_j}) = 2H \tag{9.22}$$

accordingly,

$$< H >= \frac{1}{2}fkT \tag{9.23}$$

Where f is the number of non-vanishing coefficients, namely, the number of DOF. It is valid only when the relevant of DOF can be freely excited.

9.3 Virial Theorem

The sum of products of the coordinates of the particles and the respective forces acting on them is referred to as the *virial* of the system and is denoted by the $\mathcal{V}$

$$\mathcal{V} = -3NkT \tag{9.24}$$

The relationship between the $\mathcal{V}$ and other quantities is best understood by looking at the cases of gas. In this case, the forces are the ones arising from the walls; these forces can be designated by an external pressure P that acts on the system by virtue of the fact that it is bounded by the walls. Consequently, we have a force $-PdS$ associated with an element of dS, the negative sign appears because the force is inward, while dS is outward. The Virial of the gas is given by

$$\mathcal{V}_0 = (\sum_i q_i F_i)_0 = -P \int r \cdot dS \tag{9.25}$$

where r is the position vector. By the divergence theorem,

$$\mathcal{V}_0 = -P \int (\nabla \cdot r) dV = -3PV \tag{9.26}$$

Of course, we can find

$$PV = NkT \tag{9.27}$$

$$\mathcal{V} = -2K \tag{9.28}$$

where K is the kinetic energy.

The Virial equation of states is expressed as follows

$$\frac{P}{NkT} = 1 + \frac{1}{NdkT} \left\langle \sum_{i<j} F(r_{ij}) \cdot r_{ij} \right\rangle = 1 - \frac{1}{NdkT} \left\langle \sum_{i<j} \frac{\partial u(r_{ij})}{\partial r_{ij}} r_{ij} \right\rangle \tag{9.29}$$

9.4 Homework

1. 3.5
2. 3.8
3. 3.9
4. 3.13
5. 3.15

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Lecture 10: Harmonic Oscillator

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

10.1 Harmonic oscillators

A system of N independent harmonic oscillators is representing two important problems,

1. the theory of black-body radiation
2. the theory of lattice vibrations

We start with the specialized situation when the oscillators can be treated classically. The Hamiltonian of any one of them is then given by

$$H(q_i, p_i) = \frac{1}{2}m\omega^2 q_i^2 + \frac{1}{2m}p_i^2 \tag{10.1}$$

For the single-oscillator partition function, we readily obtain

$$\begin{aligned} Q_1(\beta) &= \int\int \exp[-\beta(\frac{1}{2}m\omega^2 q^2 + \frac{1}{2m}p^2)]\frac{dqdp}{h} \\ &= \frac{1}{h}\left(\frac{2\pi}{\beta m\omega^2}\right)^{1/2}\left(\frac{2\pi m}{\beta}\right)^{1/2} \\ &= \frac{1}{\beta\hbar\omega} = \frac{kT}{\hbar\omega} \end{aligned} \tag{10.2}$$

This representing a classical counting of the average number of accessible micro states - that is, kT divided by the quantum harmonic oscillator energy spacing. The partition function of the $N-$oscillator system would be

$$Q_N(\beta) = [Q_1(\beta)]^N = (\beta\hbar\omega)^{-N} = \left(\frac{kT}{\hbar\omega}\right)^N \tag{10.3}$$

here we assume that the oscillators are distinguishable. It is actually photons in one case and phonons in the other, which distribute themselves over the various oscillator levels, that are indistinguishable.

A is now given by

$$A = -kT\ln Q_N = NkT\ln\frac{\hbar\omega}{kT} \tag{10.4}$$

And

$$\mu = kT\ln\left(\frac{\hbar\omega}{kT}\right) \tag{10.5}$$

$$P = 0 \tag{10.6}$$

$$S = Nk\left[\ln(\frac{\hbar\omega}{kT}) + 1\right] \tag{10.7}$$

$$U = NkT \tag{10.8}$$

$$C_p = C_V = Nk \tag{10.9}$$

We can also determine the density of states $g(E)$

$$g(E) = \frac{1}{(\hbar\omega)^N}\frac{1}{2\pi i}\int_{\beta'+i\infty}^{\beta'-i\infty}\frac{e^{(\beta E)}}{\beta^N}d\beta \tag{10.10}$$

$$g(E) = \begin{cases} \frac{1}{(\hbar\omega)^N}\frac{E^N-1}{(N-1)!} & \text{if } E \geq 0 \\ 0 & \text{if } E < 0 \end{cases} \tag{10.11}$$

Taking $N \ll 1$ and using the Stirling approximation, we get

$$S(N,E) = k\ln g(E) \approx NK\left[\ln\left(\frac{E}{N\hbar\omega}\right) + 1\right] \tag{10.12}$$

which gives for the temperature of the system

$$T = \left(\frac{\partial S}{\partial E}\right)_N^{-1} = \frac{E}{Nk} \tag{10.13}$$

10.2 Quantum situation

When it comes to the quantum situation,

$$\epsilon_n = (n + 1/2)\hbar\omega \tag{10.14}$$

We have for the single-oscillator partition function

$$Q_1(\beta) = \sum_{n=0}^{\infty} e^{-\beta(n+1/2)\hbar\omega} = \frac{\exp(-1/2\beta\hbar\omega)}{1-\exp(-\beta\hbar\omega)} \tag{10.15}$$

The N-oscillator partition function is then given by

$$Q_N(\beta) = [Q_1(\beta)]^N = \frac{\exp(-N/2\beta\hbar\omega)}{(1-\exp(-\beta\hbar\omega))^N} \tag{10.16}$$

Therefore, we get

$$A = NkT\ln Q_1(\beta) = NkT\ln\left[\frac{\exp(-1/2\beta\hbar\omega)}{(1-\exp(\beta\hbar\omega))}\right] = N\left[\frac{1}{2}\hbar\omega + kT\ln(1-e^{-\beta\hbar\omega})\right] \tag{10.17}$$

$$\mu = A/N = \frac{1}{2}\hbar\omega + kT\ln(1 - e^{-\beta\hbar\omega}) \tag{10.18}$$

$$P = 0 \tag{10.19}$$

$$S = Nk\left[\frac{\beta\hbar\omega}{e^{\beta\hbar\omega} - 1} - \ln(1 - e^{-\beta\hbar\omega})\right] \tag{10.20}$$

$$U = N\left[\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\beta\hbar\omega - 1}}\right] \tag{10.21}$$

$$C_P = C_V = Nk(\beta\hbar\omega)^2\frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2} \tag{10.22}$$

Clearly, the quantum mechanical oscillators do not obey the equipartition theorem. The mean energy per oscillator is different from kT. Only in the high T limit, the mean energy tend to be equal to kT.

For $kT \gg \hbar\omega$, the formula goes to classical picture.

We shall now determine the density of states $g(E)$ of the N-oscillator system from its partition function. Carrying out the binomial expansion of the expression, we have

$$Q_N(\beta) = \sum_{R=0}^{\infty}\binom{N+R-1}{R}e^{-\beta(\frac{1}{2}N\hbar\omega + R\hbar\omega)} \tag{10.23}$$

Comparing this with the formula

$$Q_N(\beta) = \int_0^{\infty} g(E)e^{-\beta E}dE \tag{10.24}$$

We conclude that

$$g(E) = \sum_{R=0}^{\infty}\binom{N+R-1}{R}\delta\left(E - [R + \frac{1}{2}N]\hbar\omega\right) \tag{10.25}$$

where $\delta(x)$ denotes the Dirac delta function. It implies that there are $(N+R-1)!/R!(N-1)!$ microstates available to the system when its energy E has the discrete value $(R + \frac{1}{2}N)\hbar\omega$, where R= 0,1,2,..., and that no micro-state is available for other values of E. This is hardly surprising, but it is instructive to look at this result from a different point of view.

We consider the problem that arises naturally in the micro-canonical ensemble theory. Given an energy E for distribution among a set of N harmonic oscillators, each of which can be in any one of the eigenstates, what is the total number of distinct ways in which the process of distribution can be carried out? In view of the form of the eigenvalues ϵ_n, it makes sense to give away, right in the beginning, the zero-point energy $\frac{1}{2}\hbar\omega$ to each of the N oscillators and convert the rest of it into quanta. Let R be the number of these quanta, then

$$R = \left(E - \frac{1}{2}N\hbar\omega\right)/\hbar\omega \tag{10.26}$$

Clearly, R must be an integer, by implication, E must be of the form $(R + \frac{1}{2}N)\hbar\omega$. The problem then reduces to determining the number of distinct ways of arranging R to N oscillators. The answer is clearly

$$\frac{(R+N-1)!}{R!(N-1)!} \tag{10.27}$$



Figure 10.1: Distribution of energies.

We can now determine the entropy of the system from the number, since $N \gg 1$,we have,

$$\begin{aligned} S &\approx k[\ln(R+N)! - \ln R! - \ln N!] \\ &\approx k[(R+N)\ln(R+N) - R\ln R - N\ln N] \end{aligned} \tag{10.28}$$

The number R is a measure of the energy E fo the system, we can calculate,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N = \left(\frac{\partial S}{\partial R}\right)_N \frac{1}{\hbar\omega} = \frac{k}{\hbar\omega}\ln\left(\frac{R+N}{R}\right) = \frac{k}{\hbar\omega}\ln\left(\frac{E+\frac{1}{2}N\hbar\omega}{E-\frac{1}{2}N\hbar\omega}\right) \tag{10.29}$$

so that,

$$\frac{E}{N} == \frac{1}{2}\hbar\omega\frac{\exp(\hbar\omega/kT)+1}{\exp(\hbar\omega/kT)-1} \tag{10.30}$$

This is consistent with our thermodynamic limit.

Finally, we may consider

$$\frac{(R+N-1)(R+N-2)...(R+1)}{(N-1)!} \approx \frac{R^N-1}{(N-1)!} \tag{10.31}$$

with

$$R \approx E/\hbar\omega \tag{10.32}$$

The corresponding expression for the entropy turns out to be

$$S \approx k[N\ln(R/N)+N] \approx Nk[\ln(\frac{E}{N\hbar\omega})+1] \tag{10.33}$$

which gives

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N \approx \frac{Nk}{E} \tag{10.34}$$

so

$$\frac{E}{N} \approx kT \tag{10.35}$$

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Lecture 11: The Grand Canonical Ensemble

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

11.1 Background

We established the basis of ensemble theory and made a detailed study of the micro-canonical and canonical ensembles. However, for a number of problems, the usefulness of them turn out to be rather limited and it appears that a further generalization of this formalism is called for. From micro-canonical to canonical ensemble, the macro-state of the systems was defined from a fixed (N, V, E) to (N, V, T). We can continue to expand them to a state where both N and E are variables.

11.2 Equilibrium between a system and a particle-energy reservoir

We consider the system A as immersed in a large reservoir A′, with which it can exchange both energy and particles. After some time has elapsed, the system and the reservoir are supposed to attain a state of mutual equilibrium. Then, they will reach a common temperature T and chemical potential μ. The fraction of the total number of particlesN^0 and the fraction of the total energy E^0 are conserved. While the system can take any energy E_r and number of particles N_r anywhere between zero and unity. Therefore, it satisfies,

$$N_r + N_r^{'} = N^0 \quad \text{const} \tag{11.1}$$

$$E_r + E_r^{'} = E^0 \quad \text{const} \tag{11.2}$$

Since the reservoir is supposed to be much larger than the system,

$$N_r \ll N_r^{'} \quad \text{const} \tag{11.3}$$

$$E_r \ll E_r^{'} \quad \text{const} \tag{11.4}$$

Now, in the manner of Lec 07, the probability $P_{r,s}$ that at any time t, the system A is found to be in an (N_r, E_s) state would be directly proportional to the number of micro-states $\Omega'(N_r', E_s')$ that the reservoir can have for the corresponding macro-state (N_r', E_s') . Thus,

$$P_{r,s} \propto \Omega'(N^0 - N_r, E^0 - E_s) \tag{11.5}$$

This can be written as

$$\begin{aligned} \ln \Omega'(N^0 - N_r, E^0 - E_s) &= \ln \Omega'(N^0, E^0) + \frac{\partial \ln \Omega'}{\partial N'}_{N'=N^0}(-N_r) + \frac{\partial \ln \Omega'}{\partial E'}_{E'=E^0}(-E_s) + ... \\ &\approx \ln \Omega'(N^0, E^0) + \frac{\mu'}{kT'} N_r + \frac{1}{kT'} E_s \end{aligned} \tag{11.6}$$

Therefore, we obtain the desired result:

$$P_{r,s} \propto \exp(-\alpha N_r - \beta E_s) \tag{11.7}$$

where

$$\alpha = \mu/kT, \quad \beta = 1/kT \tag{11.8}$$

By normalization, it becomes,

$$P_{r,s} = \frac{\exp(-\alpha N_r - \beta E_s)}{\sum_{r,s} \exp(-\alpha N_r - \beta E_s)} \tag{11.9}$$

The summation in the denominator goes over all the (N_r, E_s) states accessible to the system A.

11.3 A system in the grand canonical ensemble

We now consider an ensemble of $\mathcal{N}$ identical systems mutually sharing a total number of particles $\mathcal{N}\bar{N}$ and a total energy of $\mathcal{N}\bar{E}$. Let $n_{r,s}$ denote the number of states of the systems that have, at any time t, the number of N_r of particles and amount of E_s of energy (r, s = 0, 1, 2, ...) then follows,

$$\begin{aligned} \sum_{r,s} n_{r,s} &= \mathcal{N} \\ \sum_{r,s} n_{r,s} N_r &= \mathcal{N}\bar{N} \\ \sum_{r,s} n_{r,s} &= \mathcal{N}\bar{E} \end{aligned} \tag{11.10}$$

Any set $\{n_{r,s}\}$ of the numbers $n_{r,s}$ which satisfies the restrictive conditions, represents one of the possible modes of distribution of particles and energy among the members of our ensemble. Furthermore, any such mode of distribution can be realized in different ways, the number is

$$W\{n_{r,s}\} = \frac{\mathcal{N}!}{\prod_{r,s}(n_{r,s}!)} \tag{11.11}$$

We may now define the most probable mode of distribution $n^*_{r,s}$ as the one that have maximum W. Following the way which we did on canonical ensemble, we obtain for a large ensemble,

$$\frac{n^*_{r,s}}{\mathcal{N}} = \frac{\exp(-\alpha N_r - \beta E_s)}{\sum_{r,s} \exp(-\alpha N_r - \beta E_s)} \tag{11.12}$$

Alternatively, we may define the expectation values of the numbers $n_{r,s}$ as

$$< n_{r,s} >= \frac{\sum_{\{n_{r,s}\}} n_{r,s} W\{n_{r,s}\}}{\sum_{\{n_{r,s}\}} W\{n_{r,s}\}} \tag{11.13}$$

where the primed summations go over all distributions sets that conforms to the conditions. When the number $\mathcal{N}$ is very large, we find

$$\lim_{N\to\infty} \frac{< n_{r,s} >}{\mathcal{N}} \approx \frac{n^*_{r,s}}{\mathcal{N}} = \frac{\exp(-\alpha N_r - \beta E_s)}{\sum_{r,s} \exp(-\alpha N_r - \beta E_s)} \tag{11.14}$$

in agreement with the equation with we derived from the macroscopic picture.

11.4 Physical quantities

To establish the connection between the statistics of the grand canonical ensemble and the thermodynamics of the system, we introduce a quantity q,

$$q = \ln\big[\sum_{r,s} \exp(-\alpha N_r - \beta E_s)\big] \tag{11.15}$$

the quantity q is a function of α, β and also all the E_s. Taking the differential of q and making use of equations, we get

$$dq = -\bar{N}d\alpha - \bar{E}d\beta - \frac{\beta}{\mathcal{N}}\sum_{r,s} < n_{r,s} > dE_s \tag{11.16}$$

so that

$$d(q + \alpha\bar{N} + \beta\bar{E}) = \beta\big(\frac{\alpha}{\beta} + d\bar{E} - \frac{1}{\mathcal{N}}\sum_{r,s} < n_{r,s} > dE_s\big) \tag{11.17}$$

To interpret the terms, we recall the first law of thermodynamics,

$$\delta Q = -\mu d\bar{N} + d\bar{E} + \delta W \tag{11.18}$$

where the various symbols have their usual meanings.

$$\delta W = -\frac{1}{\mathcal{N}}\sum_{r,s} < n_{r,s} > dE_s, \;\; \mu = -\frac{\alpha}{\beta} \tag{11.19}$$

with the result that

$$d(q + \alpha\bar{N} + \beta\bar{E}) = \beta\delta Q \tag{11.20}$$

the quantity $(q + \alpha\bar{N} + \beta\bar{E})$ would be then identified with the thermodynamic variable S/k ,

$$q = \frac{S}{k} - \alpha\bar{N} - \beta\bar{E} = \frac{TS + \mu\bar{N} - \bar{E}}{kT} = \frac{G - (E - TS)}{kT} = \frac{PV}{kT} \tag{11.21}$$

To derive further results, we also defined

$$z = e^{-\alpha} = e^{\mu/kT} \tag{11.22}$$

the parameter z is referred to fugacity of the system, In terms of z, the $q-$potential takes the form

$$q = \ln\big[\sum_{r,s} z^{N_r} e^{-\beta E_s}\big] = \ln\big[\sum_{r,s} z^{N_r} Q_{N_r}(V,T)\big] \tag{11.23}$$

So we can write

$$q(z,V,T) = \ln \mathcal{Q}(z,V,T) \tag{11.24}$$

where $\mathcal{Q}$ is the grand partition function

$$\mathcal{Q}(z,V,T) = \sum_{N_r=0}^{\infty} Z^{N_r} Q_{N_r}(V,T) \tag{11.25}$$

With these quantities being defined, we can now derive the leading thermodynamic quantities,

$$P(z,V,T) = \frac{kT}{V}q(z,V,T) = \frac{kT}{V}\ln \mathcal{Q}(z,V,T) \tag{11.26}$$

$$N(z,V,T) = z\left[\frac{\partial}{\partial z}q(z,V,T)\right]_{V,T} = kT\left[\frac{\partial}{\partial \mu}q(z,V,T)\right]_{V,T} \tag{11.27}$$

$$U(z,V,T) = -\left[\frac{\partial}{\partial \beta}q(z,V,T)\right]_{V,T} = kT^2\left[\frac{\partial}{\partial T}q(z,V,T)\right]_{V,T} \tag{11.28}$$

The Helmholtz free energy can be expressed as,

$$A = N\mu - PV = NkT\ln z - kT\ln \mathcal{Q}(z,V,T) = -kT\ln\frac{\mathcal{Q}(z,V,T)}{z^N} \tag{11.29}$$

which may be compared with the canonical ensemble

$$A = -kT\ln Q(N,V,T) \tag{11.30}$$

Finally, the entropy is

$$S = \frac{U-A}{T} = kT\left(\frac{\partial q}{\partial T}\right)_{z,V} - Nk\ln z + kq \tag{11.31}$$

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Lecture 12: Examples

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

12.1 Ideal gas

For the canonical ensemble, we have the partition function as

$$Q_N(V,T) = \frac{[Q_1(V,T)]^N}{N!} \tag{12.1}$$

where $Q_1(V,T)$ is the partition function of a single molecule. Since the particles are non-localized they can be anywhere, the function Q_1 could be written as

$$Q_1(V,T) = Vf(T) \tag{12.2}$$

where $f(T)$ is a function of temperature alone. We thus obtain for the grand partition of the gas,

$$\mathcal{Q}(z,V,T) = \sum_{N_r=0}^{\infty} Z^{N_r} Q_{N_r}(V,T) = \sum_{N_r=0}^{\infty} \frac{[zVf(T)]^{N_r}}{N_r!} = \exp[zVf(T)] \tag{12.3}$$

which gives

$$q(z,V,T) = zVf(T) \tag{12.4}$$

We can then derive the following quantities,

$$\begin{aligned} P &= zkTf(T) \\ N &= zVf(T) \\ U &= zVkT^2 f'(T) \\ A &= NkT\ln z - zVkTf(T) \\ S &= NkT\ln z + zVk[Tf'(T) + f(T)] \end{aligned} \tag{12.5}$$

Again, we can also derive $PV = NkT$ from the above results. Considering N and U, we get

$$U = NkT^2 f'(T)/f(T) \tag{12.6}$$

which gives,

$$C_V = Nk\frac{2Tf(T)f'(T) + T^2[f(T)f''(T) - [f'(T)]^2]}{[f(T)]^2} \tag{12.7}$$

In simple cases, the function $f(T)$ turns out to be directly proportional to a certain power of T. Assuming $f(T) \propto T^n$, equations become

$$U = n(NkT) \tag{12.8}$$

$$C_V = n(Nk) \tag{12.9}$$

Accordingly, the pressure in such cases is directly proportional to the energy density of the gas.

Finally, eliminating z between N, A, S, we obtain A and S as functions of N, V, T.

12.2 Harmonic oscillator

The microscopic entities constituting the system are mutually indistinguishable. The partition function $Q_N(V,T)$ of such a system is:

$$Q_N(V,T) = [Q_1(V,T)]^N \tag{12.10}$$

At the same time, the single partition function $Q_1(V,T)$ is independent of the volume, so we write

$$Q_1(V,T) = \phi(T) \tag{12.11}$$

where $\phi(T)$ is a function of temperature alone. We then obtain the grand partition function of the system,

$$\mathcal{Q}(z,V,T) = \sum_{N_r=0}^{\infty} [z\phi(T)]^{N_r} = [1 - z\phi(T)]^{-1} \tag{12.12}$$

The thermodynamics of the system follows straightforwardly,

$$\begin{aligned} P &= \frac{kT}{V} q(z,T) = -\frac{kT}{V} \ln[1 - z\phi(T)] \\ N &= \frac{z\phi(T)}{1 - z\phi(T)} \\ U &= \frac{zkT^2\phi(T)}{1 - z\phi(T)} \\ A &= NkT \ln z + kT \ln[1 - z\phi(T)] \\ S &= -Nk \ln z - k \ln[1 - z\phi(T)] + \frac{zkT\phi'(T)}{1 - z\phi(T)} \end{aligned} \tag{12.13}$$

From N, we get

$$z\phi(T) = \frac{N}{N+1} \approx 1 - \frac{1}{N} \quad (N \gg 1) \tag{12.14}$$

It follows that

$$1 - z\phi(T) = \frac{1}{N+1} \approx \frac{1}{N} \tag{12.15}$$

So

$$U/N = kT^2\phi'(T)/\phi(T) \tag{12.16}$$

$$A/N = -kT \ln \phi(T) + O\left(\frac{\ln N}{N}\right) \tag{12.17}$$

substituting

$$\phi(T) = [2 \sinh(\hbar\omega/2kT)]^{-1} \tag{12.18}$$

into these formulas, we obtain the results of quantum mechanical 1D harmonic oscillators.

substituting

$$\phi(T) = kT/\hbar\omega \tag{12.19}$$

we get the results for classical 1D harmonic oscillators.

12.3 Density and energy fluctuation

How will the thermodynamics vary from different ensembles? The true answer turns out that the differences might not be that much, if the fluctuation of these variables is small (practically negligible). Therefore, it is useful to evaluate the fluctuations.

Recalling that

$$\overline{N} = \frac{\sum_{r,s} N_r e^{-\alpha N_r - \beta E_s}}{\sum_{r,s} e^{-\alpha N_r - \beta E_s}} \tag{12.20}$$

It follows that,

$$\left(\frac{\partial \overline{N}}{\partial \alpha}\right)_{\beta, E_s} = -\overline{N^2} + \overline{N}^2 \tag{12.21}$$

Thus,

$$\overline{(\Delta N)^2} = \overline{N^2} - \overline{N}^2 = \left(\frac{\partial \overline{N}}{\partial \alpha}\right)_{T,V} = kT\left(\frac{\partial \overline{N}}{\partial \mu}\right)_{T,V} \tag{12.22}$$

The mean square fluctuation in the particle density $n = (N/V)$

$$\frac{\overline{(\Delta n)^2}}{\overline{n}^2} = \frac{\overline{(\Delta N)^2}}{\overline{N}^2} = \frac{kT}{\overline{N}^2}\left(\frac{\partial \overline{N}}{\partial \mu}\right)_{T,V} \tag{12.23}$$

In terms of the variable $v = V/\overline{N}$, we write

$$\frac{\overline{(\Delta n)^2}}{\overline{n}^2} = \frac{kTv^2}{V^2}\left(\frac{\partial (V/v)}{\partial \mu}\right)_{T,V} = \frac{kT}{V}\left(\frac{\partial v}{\partial \mu}\right)_{T} \tag{12.24}$$

Recall the thermodynamic relation

$$d\mu = vdP - sdT \tag{12.25}$$

we get

$$\frac{\overline{(\Delta n)^2}}{\overline{n}^2} = \frac{kT}{V}\frac{1}{v}\left(\frac{\partial v}{\partial \mu}\right)_{T} = \frac{kT}{V}\kappa_T \tag{12.26}$$

where κ_T is the isothermal compressibility of the system. Thus, the rms of particle density is $O(N^{-1/2})$ and hence negligible. However, there are some exceptions. When it comes to the phase transitions, the compressibility of a given system can be very large!

We shall now examine the fluctuation in the energy of the system.

$$\overline{(\Delta E)^2} = \bar{E^2} - \bar{E}^2 = -\left(\frac{\partial \bar{E}}{\partial \beta}\right)_{z,V} = kT^2\left(\frac{\partial U}{\partial T}\right)_{z,V} \tag{12.27}$$

To make it more comprehensible,

$$\left(\frac{\partial U}{\partial T}\right)_{N,V} + \left(\frac{\partial U}{\partial N}\right)_{T,V}\left(\frac{\partial N}{\partial T}\right)_{z,V} \tag{12.28}$$

now, in view of the fact that

$$N = -\left(\frac{\partial}{\partial \alpha}\ln \mathcal{Q}\right)_{\beta,V}, \quad U = -\left(\frac{\partial}{\partial \beta}\ln \mathcal{Q}\right)_{\alpha,V} \tag{12.29}$$

we have

$$\left(\frac{\partial N}{\partial \beta}\right)_{\alpha,V} = \left(\frac{\partial N}{\partial \alpha}\right)_{\beta,V} \tag{12.30}$$

and

$$\left(\frac{\partial N}{\partial T}\right)_{z,V} = \frac{1}{T}\left(\frac{\partial U}{\partial \mu}\right)_{T,V} \tag{12.31}$$

substituting it, we get

$$\overline{(\Delta E)^2} = kT^2 C_V + kT\left(\frac{\partial U}{\partial N}\right)_{T,V}\left(\frac{\partial U}{\partial \mu}\right)_{T,V} \tag{12.32}$$

Finally,

$$\overline{(\Delta E)^2} = \overline{(\Delta E)^2}_{\text{canonical}} + \left[\frac{\partial U}{\partial N}\right]^2_{T,V} \overline{(\Delta N)^2} \tag{12.33}$$

It ends up with the picture that the value of canonical ensemble plus a contribution due to particle fluctuation.

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Lecture 13: Phase diagrams

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

13.1 Phase diagrams

One of the great successes of thermodynamics and statistical mechanics has been in the study of phase transitions. Condensed materials exist in a variety of phases that depend on thermodynamic parameters such as P, V, T, H, E. Thermodynamics and statistical mechanics can be used to determine the properties of individual phases, and the locations and the characteristics of the phase transitions.

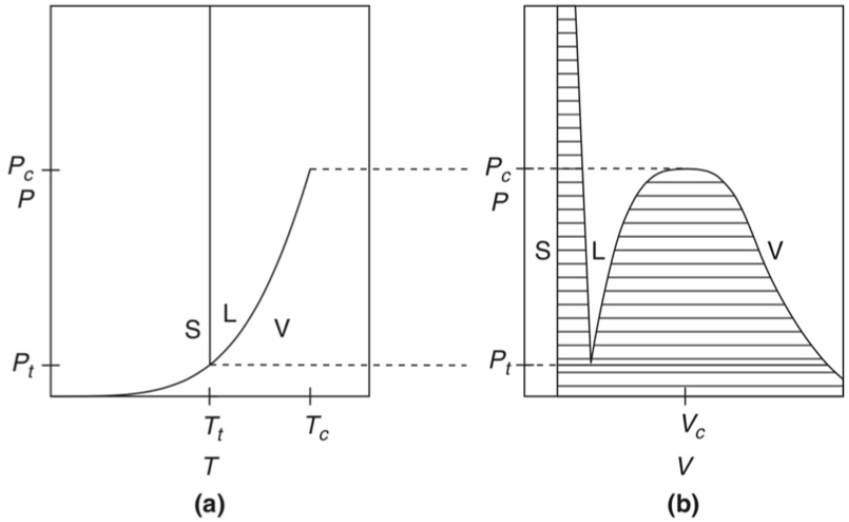


Figure 13.1: $P-T$ and $P-V$ phase Argon.

1. solid, liquid, vapor and supercritical phases
2. critical point
3. triple point
4. coexisting line
5. first-order phase transition
6. discontinuity
7. continuous phase transition

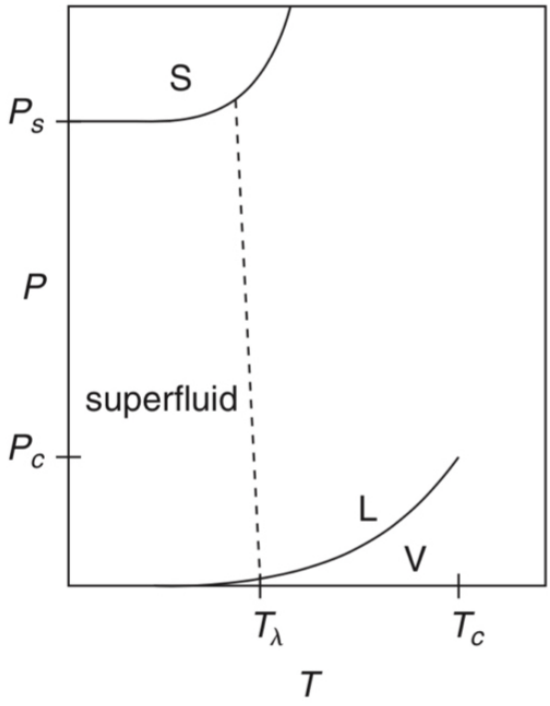


Figure 13.2: $P-T$ phase diagram of He.

1. quantum regime
2. superfluid/supersolid at low $P-T$

13.2 Clausius-Clapeyron equation

The Gibbs free energy of the phases of a material determines the phase diagram.

$$G(N,P,T) = U - TS + PV = A + PV = \mu N \tag{13.1}$$

As the number of molecules in each phase changes, the Gibbs free energy changes by an amount

$$dG = (\frac{\partial G_A}{\partial N_A})_{T,P} dN_A + (\frac{\partial G_B}{\partial N_B})_{T,P} dN_B = (\mu_A - \mu_B) dN_A \tag{13.2}$$

where dN_A is the change of number of particles.

At the equilibrium, G reaches its minimum, so $dG \leq 0$. If $\mu_A > \mu_B$, the number of molecules in phase B will increase and the number in phase A will decrease as the system approaches equilibrium. If $\mu_A < \mu_B$, the number of molecules in phase A will increase and B will decrease. When $\mu_A = \mu_B$, they reach equilibrium.

Let's consider the example of water,

$$\mu_A(P_\delta(T), T) = \mu_B(P_\delta(T), T)) \tag{13.3}$$

The derivatives of the chemical potentials are related by

$$\left(\frac{\partial \mu_A}{\partial T}\right)_P + \left(\frac{\partial \mu_A}{\partial P}\right)_T \frac{dP_\delta}{dT} = \left(\frac{\partial \mu_B}{\partial T}\right)_P + \left(\frac{\partial \mu_B}{\partial P}\right)_T \frac{dP_\delta}{dT} \tag{13.4}$$

The entropy per particle $s = S/N$ and specific volume $v = V/N$ are given by

$$s = -\left(\frac{\partial \mu}{\partial T}\right)_P v = -\left(\frac{\partial \mu}{\partial P}\right)_T \tag{13.5}$$

Therefore,

$$\frac{dP}{dT} = \frac{s_B - s_A}{v_B - v_A} = \frac{\Delta s}{\Delta v} = \frac{L}{T\Delta v} \tag{13.6}$$

Where $L = T\Delta s$ is the latent heat per particle. The slope of the coexistence curve depends on the discontinuities of the entropy per particle and the volume per particle.

At a triple point, the chemical potentials of three phases are equal,

$$\mu_A = \mu_B = \mu_C \tag{13.7}$$

The slopes of the three coexistence lines that define the triple point are related since $\Delta s_{AB} + \Delta s_{BC} + \Delta s_{CA} = 0$ and $\Delta v_{AB} + \Delta v_{BC} + \Delta v_{CA} = 0$. This ensures that each coexistence line between two phases at the triple point points into the third phase. Problem. 4.17

13.3 Quiz

Sketch the $P - T$ and $P - V$ phase diagrams of water, taking into account that the mass density of liquid water is greater than that of solid ice.

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Lecture 14: Interacting systems: Cluster expansion

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

14.1 Cluster expansion

So far, we considered only systems composed of *noninteracting* entities. The results apply to only a few systems. For a contact between the theory and experiment, one must take into account the interactions between particles. We briefly talked about the possibility of including interactions when we introduced the *virial theorem*. In this chapter, we will consider to derive a general approach to handle such systems by the forms of series expansions, which was developed by Mayer.

14.2 Cluster expansion for a classical gas

Consider a monoatomic gas, its Hamiltonian is given by

$$H = \sum_i \left(\frac{p_i^2}{2m}\right) + \sum^{\text{all pairs}} u_{ij} \tag{14.1}$$

The summation in the 2nd part goes over all $N(N-1)/2$ pairs of N particles. In general, the potential u_{ij} is function of the relative position vectors r_{ij}.

The partition function is thus given by

$$\begin{aligned} Q_N(V,T) &= \frac{1}{N!h^{3N}} \int \exp\left\{ -\beta \sum_i \left(\frac{p_i^2}{2m}\right) - \beta \sum u_{ij} \right\} d^{3N}p d^{3N}r \\ &= \frac{1}{N!\lambda^{3N}} \int \exp\left[-\beta \sum u_{ij} \right] d^{3N}r \\ &= \frac{1}{N!\lambda^{3N}} Z_N(V,T) \end{aligned} \tag{14.2}$$

where $\lambda = h/(2\pi mkT)^{1/2}$ is the mean thermal length of the particles due to its kinetic energy, while $Z_N(V,T)$ is

$$Z_N(V,T) = \int \exp\left[-\beta \sum u_{ij} \right] d^{3N}r = \int \prod_{i<j} \exp(-\beta u_{ij}) d^{3N}r \tag{14.3}$$

Clearly, $Z_N(V,T)$ is related to the configurational integral of the system. If it is ideal gas,

$$Z_N(V,T) = V^N \tag{14.4}$$

To treat the nonideal case, Mayer introduced the two particle function f_{ij},

$$f_{ij} = \exp(-\beta u_{ij}) - 1 \tag{14.5}$$

Since u is a function of r_{ij}, we expect that u will have large repulsion within short distance, and go to zero at long range. So f will be -1 with in short distance, and 0 at long range.

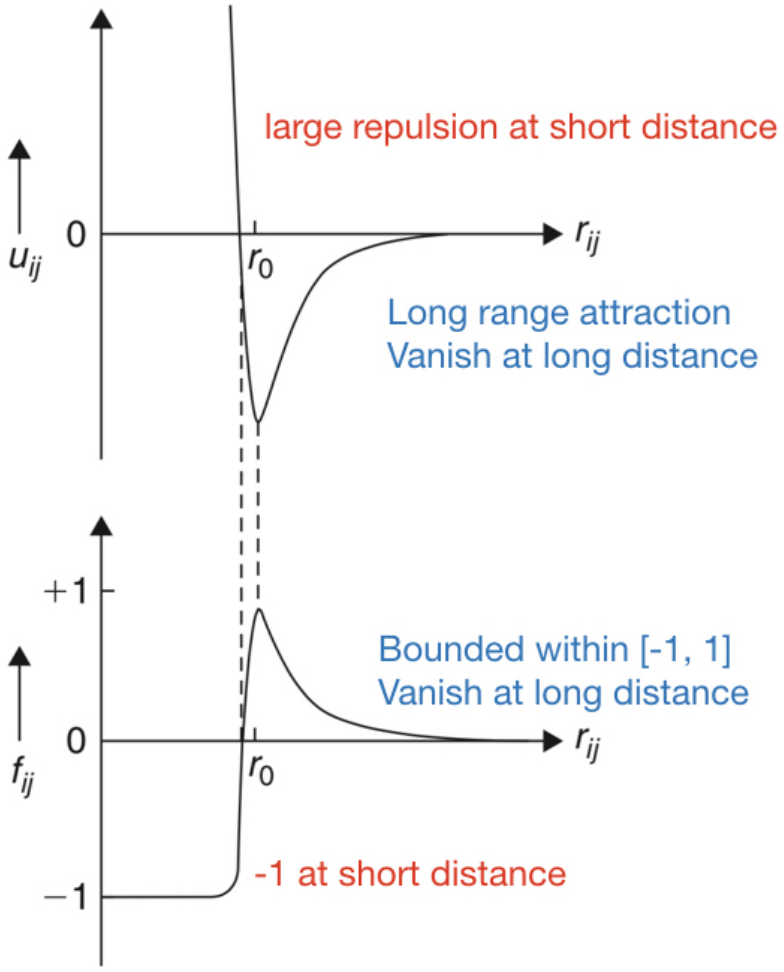


Figure 14.1: A typical plot of the two-body potential function u_{ij} and the corresponding Mayer function f_{ij}.

Now, to evaluate the integral, we expand the integrand in ascending powers of the functions f_{ij}:

$$\begin{aligned} Z_N(V,T) &= \int \prod_{i<j} (1 + f_{ij}) d^3 r_1 \cdots d^3 r_N \\ &= \int \left[1 + \sum f_{ij} + \sum f_{ij} f_{kl} + \cdots \right] d^3 r_1 \cdots d^3 r_N \end{aligned} \tag{14.6}$$

Where you can regard the expansions represent the 2-body, 3-body, ... and N-body clusters.

N-particle graph, is a collection of N distinct circles, numbered 1,2,...,N, with a number of lines linking

some (or all) of the circles.

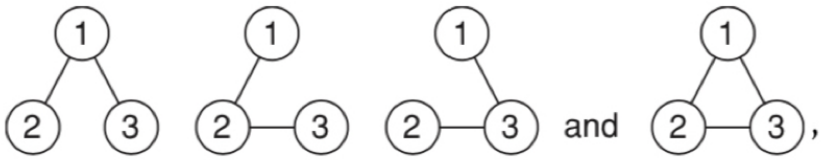
$$\int (f_\alpha f_\beta \cdots f_\lambda) d^3r_1 \cdots d^3r_N \tag{14.7}$$

A graph having the same number of linked pairs as this one but with the set $(\alpha', \beta', \cdots, \lambda')$ distinct from the set $(\alpha, \beta, \cdots, \lambda)$ will be counted as a distinct graph,

Therefore, we have

$$Z_N(V,T) = \text{sum of all distinct } N\text{-particle graphs} \tag{14.8}$$

Also, we note that some graphs have degeneracy while some not. For instance, a group of three particles leads to four 3-clusters,



of which the first three are equal. Therefore, we introduce the notation of $b_l(V,T)$

$$b_l(V,T) = \frac{1}{l!\lambda^{3(l-1)V}} \times (\text{the sum of all possible } l\text{-clusters}) \tag{14.9}$$

Some of the simple cluster integrals are:

$$\begin{aligned}
b_1 &= \frac{1}{V}\int d^3r_1 = 1 \\
b_2 &= \frac{1}{2\lambda^3 V}\iint f_{12} d^3r_1 d^3r_2 \\
&\approx \frac{1}{2\lambda^3}\int f_{12} d^3r_{12} = \frac{2\pi}{\lambda^3}\int f(r) r^2 dr \\
&= \frac{2\pi}{\lambda^3}\int (\exp(-u(r)/kT) - 1) r^2 dr \\
b_3 &= \frac{1}{6\lambda^6 V}\iiint (f_{12}f_{13} + f_{12}f_{23} + f_{13}f_{23} + f_{12}f_{13}f_{23}) d^3r_1 d^3r_2 d^3r_3 \\
&\approx \frac{1}{6\lambda^6 V}\left[3V\iint f_{12}f_{13} d^3r_{12} d^3r_{13} + V\iint f_{12}f_{13}f_{23} d^3r_{12} d^3r_{13}\right] \\
&= 2b_2^2 + \frac{1}{6\lambda^6}\iint f_{12}f_{13}f_{23} d^3r_{12} d^3r_{13}
\end{aligned} \tag{14.10}$$

Note that f_{ij} drops rapidly to zero as r_{ij} becomes large, so that, if the position of i does not happen to be within the molecular distance of the walls of the vessel,

$$\int f_{ij} dr_j^3 = \int_0^\infty f(r) 4\pi r^2 dr = \beta \tag{14.11}$$

$$b_2 = \int \beta dr_i = \beta V \tag{14.12}$$

Obviously, an N-particle graph will have a number of clusters, say m_1 are 1-clusters, m_2 are 2-clusters, and so on. The numbers $\{m_l\}$ must satisfy

$$\sum_{l=1}^{N} l m_l = N \tag{14.13}$$

However, a given set of numbers of $\{m_l\}$ does not specify a unique, single graph. it represents a collection of graphs the sum total of which may be denoted by the symbol $S\{m_l\}$. We may then write

$$Z_N(V,T) = \sum_{\{m_l\}} S\{m_l\} \tag{14.14}$$

To evaluate the sum $S\{m_l\}$. We use two observations,

- There are many different ways to assign N particles of the system to the $\sum_l m_l$ clusters
- For any given assignments, many ways to form the clusters, e.g., four ways to form a 3-cluster.

from (1), we obtain a straightforward factor of

$$\frac{N!}{(1!)^{m_l}(2!)^{m_2}\cdots} = \frac{N!}{\prod_l (l!)^{m_l}} \tag{14.15}$$

from (2), there are two steps of considerations.

1. for all l-clusters were unique, there is a power of m_l
2. for any two clusters with the same number, there should be a factor of $(m_l!)$

with the help of b_l,

$$\prod_l \left[(b_l l! \lambda^{3(l-1)} V)^{m_l} / m_l! \right] \tag{14.16}$$

Therefore, we obtain the configuration integral

$$Z_N(V,T) = N! \lambda^{3N} \sum_{\{m_l\}}{}' \left[\prod_l \left\{ \left(b_l \frac{V}{\lambda^3} \right)^{m_l} \frac{1}{m_l!} \right\} \right] \tag{14.17}$$

There is a relation

$$\prod_l (\lambda^{3l})^{m_l} = \lambda^{3\sum_l l m_l} = \lambda^{3N} \tag{14.18}$$

So we express the partition function

$$Q_N(V,T) = \sum_{\{m_l\}}{}' \left[\prod_l \left\{ \left(b_l \frac{V}{\lambda^3} \right)^{m_l} \frac{1}{m_l!} \right\} \right] \tag{14.19}$$

For the grand partition function of the system,

$$\mathcal{Q}(z, V, T) = \sum_{N=0}^{\infty} z^N Q_N(V, T) \tag{14.20}$$

Writing

$$z^N = z^{\sum_l l m_l} = \prod_l (z^l)^{m_l} \tag{14.21}$$

substituting for $Q_N(V, T)$ from Eq. 14.19

$$\begin{aligned}
\mathcal{Q}(z, V, T) &= \sum_{m_1, m_2, \cdots = 0}^{\infty} \left[\prod_{l=1}^{\infty} \left\{ \left(b_l z^l \frac{V}{\lambda^3} \right)^{m_l} \frac{1}{m_l!} \right\} \right] \\
&= \prod_{l=1}^{\infty} \left[\sum_{m_l=0}^{\infty} \left\{ \left(b_l z^l \frac{V}{\lambda^3} \right) \frac{1}{m_l!} \right\} \right] \\
&= \prod_{l=1}^{\infty} \left[\exp\left(b_l z^l \frac{V}{\lambda^3} \right) \right] \\
&= \exp\left[\sum_{l=1}^{\infty} b_l z^l \frac{V}{\lambda^3} \right]
\end{aligned} \tag{14.22}$$

and hence,

$$\frac{1}{V} \ln \mathcal{Q} = \frac{1}{\lambda^3} \sum_{l=1}^{\infty} b_l z^l \tag{14.23}$$

In the limit V goes to ∞,

$$\frac{P}{kT} = \lim_{V \to \infty} \left(\frac{1}{V} \ln \mathcal{Q} \right) = \frac{1}{\lambda^3} \sum_{l=1}^{\infty} b_l z^l \tag{14.24}$$

and

$$\frac{N}{V} = \lim_{V \to \infty} \left(\frac{z}{V} \frac{\partial \ln \mathcal{Q}}{\partial z} \right) = \frac{1}{\lambda^3} \sum_{l=1}^{\infty} l b_l z^l \tag{14.25}$$

The above two equations constitute the famous cluster expansions of the Mayer-Ursell formalism. Eliminating the fugacity z, we obtain the equation of states of the system.

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Lecture 15: Virial expansion

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

15.1 Cluster expansion

The approach developed in the preceding section leads to exact results only if we apply it to the gaseous phase alone. If we aim to extend it to the phenomena of condensed phase, the critical point, we encounter serious difficulties relating to

- summation over l,
- volume dependence of the cluster integral b_l

Let's focus on the gas phase at the moment.

From

$$\frac{N}{V} = \lim_{V\to\infty}\left(\frac{z}{V}\frac{\partial \ln \mathcal{Q}}{\partial z}\right) = \frac{1}{\lambda^3}\sum_{l=1}^{\infty} l b_l z^l \tag{15.1}$$

$$\sum_{l=1}^{N} \frac{v l b_l z^l}{\lambda^3} = 1 \tag{15.2}$$

where $v = V/N$ denotes the volume per particle in the system.

For the ideal gas, we only care about the first term, thus

$$z = \frac{\lambda^3}{v} \quad \text{(ideal gas)} \tag{15.3}$$

Check Eq. 1.5.7 in the textbook.

For the real gas, as v approaches infinity, the limiting value λ^3/v is obtained for z, shows that if z is expanded as a power series in the volume only inverse power series.

$$z = a_1\frac{\lambda^3}{v} + a_2\left(\frac{\lambda^3}{v}\right)^2 + a_3\left(\frac{\lambda^3}{v}\right)^3 + a_4\left(\frac{\lambda^3}{v}\right)^4 \cdots \tag{15.4}$$

Therefore,

$$\begin{aligned} z^2 &= a_1^2\frac{\lambda^6}{v^2} + 2a_1a_2\frac{\lambda^9}{v^3} + (2a_1a_3 + a_2^2)\frac{\lambda^{12}}{v^4}\cdots \\ z^3 &= a_1^3\frac{\lambda^9}{v^3} + 3a_1^2a_2\frac{\lambda^{12}}{v^4} + \cdots \\ z^4 &= a_1^4\frac{\lambda^{12}}{v^4} + \cdots \end{aligned} \tag{15.5}$$

Now we apply Eq. 15.2,

$$\begin{aligned}
1 &\approx \sum_{l=1}^{4} \frac{v l b_l z^l}{\lambda^3} = \frac{v}{\lambda^3}(b_1 z + 2b_2 z^2 + 3b_3 z^3 + 4b_4 z^4) \\
&\approx \frac{v}{\lambda^3}\left[b_1 a_1 \frac{\lambda^3}{v} + b_1 a_2 \frac{\lambda^6}{v^2} + b_1 a_3 \frac{\lambda^9}{v^3} + b_1 a_4 \frac{\lambda^{12}}{v^4} + 2b_2 a_1^2 \frac{\lambda^6}{v^2} + 4b_2 a_1 a_2 \frac{\lambda^9}{v^3} + (4b_2 a_1 a_3 + 2b_2 a_2^2)\frac{\lambda^{12}}{v^4} \right. \\
&\quad \left. + 3b_3 a_1^3 \frac{\lambda^9}{v^3} + 9b_3 a_1^2 a_2 \frac{\lambda^{12}}{v^4} + 4b_4 a_1^4 \frac{\lambda^{12}}{v^4} \right] \\
&= a_1 b_1 + (a_2 b_1 + 2a_1^2 b_2)\frac{\lambda^3}{v} + (a_3 b_1 + 4a_1 a_2 b_2 + 3a_1^3 b_3)\frac{\lambda^6}{v^2} + (a_4 b_1 + 4b_2 a_1 a_3 + 2b_2 a_2^2 + 9a_1^2 a_2 b_3 + 4a_1^4 b_4)\frac{\lambda^9}{v^3}
\end{aligned} \tag{15.6}$$

To satisfy the condition, we obtain the following relation,

$$\begin{aligned}
a_1 b_1 = 1 &\rightarrow a_1 = b_1 = 1 \\
a_2 b_1 + 2a_1^2 b_2 = 0 &\rightarrow a_2 = -2b_2 \\
a_3 b_1 + 4a_1 a_2 b_2 + 3a_1^3 b_3 &\rightarrow a_3 = 8b_2^2 - 3b_3 \\
a_4 b_1 + 2b_2 a_1 a_3 + b_2 a_2^2 + a_1^2 a_2 b_3 + a_1^4 b_4 &\rightarrow a_4 = -40b_2^3 + 30 b_2 b_3 - 4b_4
\end{aligned} \tag{15.7}$$

We now proceed to expand Pv/kT as a series of z and then v,

$$\begin{aligned}
\frac{Pv}{kT} &= \frac{v}{\lambda^3}\sum_{l=1}^{\infty} b_l z^l \\
&\approx a_1 b_1 + (b_1 a_2 + b_2 a_1^2)\frac{\lambda^3}{v} + (b_1 a_3 + 2a_1 a_2 b_2 + a_1^3 b_3)\frac{\lambda^6}{v^3} + \cdots \\
&= b_1 - b_2 \frac{\lambda^3}{v} + (4b_2^2 - 2b_3)\frac{\lambda^6}{v^3} + \cdots \\
&= \sum_{l=1}^{\infty} a'_l(T)\left(\frac{\lambda^3}{v}\right)^{l-1}
\end{aligned} \tag{15.8}$$

This is called *virial expansion* and a_l, b_l are called *virial coefficients*. To determine the relationship between a'_l and b_l, we obtain

$$\begin{aligned}
a'_1 &= b_1 = 1 \\
a'_2 &= -b_2 = -\frac{2\pi}{\lambda^3}\int_0^{\infty}\left(e^{-u(r)/kT} - 1\right) r^2 dr \\
a'_3 &= 4b_2^2 - 2b_3 = -\frac{1}{3\lambda^6}\int\int f_{12} f_{13} f_{23} d^3 r_{12} d^3 r_{13}
\end{aligned} \tag{15.9}$$

Here we note that a'_l is completely determined by the quantities $b_1, b_2, ...b_l$, that is, by the sequence of configuration integrals $Z_1, Z_2, ...Z_l$.

From the above equations, we can find that a'_3 is completely determined by the 3-cluster [1-2-3-1]. This suggests that the higher-ordder virial may be only determined by a special subgroup of l-clusters.

$$a'_l = -\frac{l-1}{l}\beta_{l-1} \quad (l \geq 2), \tag{15.10}$$

where β_{l-1} is the so called irreducible cluster integral, defined as

$$\beta_{l-1} = \frac{1}{(l-1)!\lambda^{3(l-1)}V} \times \text{the sum of all irreducibel } l\text{-clusters} \tag{15.11}$$

Following this, we obtain

$$a'_3 = -\frac{2}{3}\beta_2 \tag{15.12}$$

The quantities β_{l-1} is dimensionless, and closely related to b_{l-1}.

15.2 2nd virial coefficients with the hard sphere potential

If a given system does not depart much from the ideal-gas behavior, its equation of state is given adequately by the first few virial coefficients. Now we can consider the explicit expression of high order terms. For a_2,

$$a'_2 = -b_2 = \frac{2\pi}{\lambda^3}\int_0^{-\infty}(1 - e^{-u(r)/kT})r^2 dr \tag{15.13}$$

Now we consider that $u(r)$ takes a typical semi-empirical formula,

$$u(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] \tag{15.14}$$

An analysis on $u(r)$ yield that it reaches a minimum of ϵ at a distance $r_0 = 2^{1/6}\sigma$ and rises to an infinitely large for $r < \sigma$, and vanishes to a small (negative) value for$r$,

$$u(r) = +\infty \quad (r < r_0) \tag{15.15}$$

$$u(r) = -u_0(r_0/r^6) \quad (r \geq r_0) \tag{15.16}$$

substituting 15.15 and 15.16 into 15.13, we obtain for the second virial coefficient

$$a'_2 = \frac{2\pi}{\lambda^3}\left[\int_0^{r_0} r^2 dr + \int_{r0}^{\infty}\left[1 - \exp\left(\frac{u_0}{kT}\left(\frac{r_0}{r}\right)^6\right)\right]dr\right] \tag{15.17}$$

The first integral is straightforward; the second one is considerably simplified if we assume that $(u_0/kT) \ll 1$, which makes the integrand very nearly equal to $(u0/kT)(r_0/r)^6$.

$$a'_2 \approx \frac{2\pi r_0^3}{3\lambda}\left(1 - \frac{u_0}{kT}\right) \tag{15.18}$$

Substituting it into the expansion 15.8, we obtain a first-order improvement on the ideal-gas law, namely

$$\begin{aligned} P &\approx \frac{kT}{v}\left\{1 + \frac{2\pi r_0^3}{3v}\left(1 - \frac{u_0}{kT}\right)\right\} \\ &= \frac{kT}{v}\left\{1 + \frac{B_2(T)}{v}\right\} \end{aligned} \tag{15.19}$$

The coefficient B_2, which is also sometimes referred to as the second virial coefficient of the system, is given by

$$B_2 = a'_2\lambda^3 \approx \frac{2\pi r_0^3}{3}\left(1 - \frac{u_0}{kT}\right) \tag{15.20}$$

In our derivation it was explicitly assumed that (i) the potential function $u(r)$ is given by the simplified expressions (3) and (4), and (ii) $(u_0/kT) \ll 1$. We cannot, therefore, expect this formula to be a faithful representation of the second virial coefficient of a real gas. Nevertheless, it does correspond, almost exactly, to the van der Waals approximation to the equation of state of a real gas. This can be seen by rewriting the expression of P in the form

$$\left(P + \frac{2\pi r_0^3 u_0}{3v^2}\right) \approx \frac{kT}{v}\left(1 + \frac{2\pi r_0^3}{3v}\right) \approx \frac{kT}{v}\left(1 - \frac{2\pi r_0^3}{3v}\right)^{-1} \tag{15.21}$$

which lead to

$$\left(P + \frac{a}{v^2}\right)(v - b) \approx kT \tag{15.22}$$

where

$$a = \frac{2\pi r_0^3 u_0}{3} \text{ and } b = \frac{2\pi r_0^3}{3} = 4v_0 \tag{15.23}$$

15.3 B_2 with Lennard Jones potential

A realistic study of the second virial coefficient requires the use of a realistic potential, such as the one given by Lennard-Jones, for evaluating the integral in (1). This has indeed been done and the results obtained are shown in Figure 10.2, where the reduced coefficient $B_2^{'}(= B_2/r_0^3)$ is plotted against the reduced temperature $T'(= kT/\epsilon)$:

$$B_2^{'}(T') = 2\pi \int_0^\infty (1 - e^{-u'(r')/T'} r'^2 dr') \tag{15.24}$$

with

$$u'(r') = \left\{\left(\frac{1}{r'}\right)^{12} - 2\left(\frac{1}{r'}\right)^{6}\right\} \tag{15.25}$$

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Lecture 16: Van der Waals equation from the Perturbation theory

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

16.1 Perturbation Theory

Recall that the configuration partition function:

$$Z_N = \int \exp(-\beta U) dr_1 \cdots dr_N \tag{16.1}$$

Now we divide U to two source of contributions

$$U = U_0 + U_1, \tag{16.2}$$

where U_1 is small compared to U_0.

We first compute Z due to U_0,

$$Z_N^0 = \int \exp(-\beta U_0) dr_1 \cdots dr_N \tag{16.3}$$

Then, we express Z_N as

$$Z_N = \int \exp(-\beta U) dr_1 \cdots dr_N = Z_N^0 \langle \exp(-\beta U_1) \rangle_0 \tag{16.4}$$

The total partition function is:

$$Q(N, V, T) = \frac{Z(N, V, T)}{N! \lambda^{3N}} \tag{16.5}$$

The free energy is given by

$$A(N, V, T) = -\frac{1}{\beta} \ln \left(\frac{Z_N}{N! \lambda^{3N}} \right) = -\frac{1}{\beta} \left(\frac{Z_N^0}{N! \lambda^{3N}} \right) - \frac{1}{\beta} \ln \langle \exp(-\beta U_1) \rangle_0 \tag{16.6}$$

We hence separate A into two terms

$$\begin{aligned} A^0(N, V, T) &= -\frac{1}{\beta} \left(\frac{Z_N^0}{N! \lambda^{3N}} \right) \\ A^1(N, V, T) &= -\frac{1}{\beta} \ln \langle \exp(-\beta U_1) \rangle_0 = -\frac{1}{\beta} \ln \sum_{k=0}^{\infty} \frac{(-\beta)^k}{k!} \langle U_1^k \rangle_0 \end{aligned} \tag{16.7}$$

Where, the 2nd term in the 2nd expression includes a power series expansion of exponential function.

Here we hope to expand A^1 in a general polynomial form,

$$A^1 = \sum_{k=1}^{\infty} \frac{(-\beta)^{k-1}}{k!} w_k \tag{16.8}$$

where w_k are a set of coefficients to be determined.

Using the fact that

$$\ln(1+x) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k} \tag{16.9}$$

We have that

$$\begin{aligned} -\frac{1}{\beta} \ln\left(\sum_{k=0}^{\infty} \frac{(-\beta)^k}{k!} \langle U_1^k \rangle_0 \right) &= -\frac{1}{\beta} \ln\left(1 + \sum_{k=1}^{\infty} \frac{(-\beta)^k}{k!} \langle U_1^k \rangle_0 \right) \\ &= -\frac{1}{\beta} \sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k} \left(\sum_{l=1}^{\infty} \frac{(-\beta)^l}{l!} \langle U_1^l \rangle_0 \right)^k \end{aligned} \tag{16.10}$$

Comparing with Eq. 16.8

$$-\frac{1}{\beta} \sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k} \left(\sum_{l=1}^{\infty} \frac{(-\beta)^l}{l!} \langle U_1^l \rangle_0 \right)^k = \sum_{k=1}^{\infty} \frac{(-\beta)^{k-1}}{k!} w_k \tag{16.11}$$

From the β^1 term, we find

$$-\frac{\beta \langle U_1 \rangle_0}{1!} = -\frac{\beta w_1}{1!} \tag{16.12}$$

so

$$w_1 = \langle U_1 \rangle_0 \tag{16.13}$$

Likewise, we get

$$w_2 = \langle U_1^2 \rangle_0 \langle U_1 \rangle_0^2 \tag{16.14}$$

$$w_3 = \langle U_1^3 \rangle_0 + 2\langle U_1 \rangle_0^3 - 3\langle U_1 \rangle_0 \langle U_1^2 \rangle_0 \tag{16.15}$$

Therefore, the free energy can be expressed as

$$\begin{aligned} A &= A^0 + w_1 - \frac{\beta}{2} w_2 + \frac{\beta^2}{6} w_3 \cdots \\ &= -\frac{1}{\beta} \ln\left(\frac{Z_N^0}{N! \lambda^{3N}} \right) + \langle U_1 \rangle_0 - \frac{\beta}{2} (\langle U_1^2 \rangle_0 \langle U_1 \rangle_0^2) + \frac{\beta^2}{6} (\langle U_1^3 \rangle_0 + 2\langle U_1 \rangle_0^3 - 3\langle U_1 \rangle_0 \langle U_1^2 \rangle_0) \cdots \end{aligned} \tag{16.16}$$

Now the next task is to obtain some analytical formula for $\langle U_1 \rangle_0$, suppose that U_1 adopts a pair potential

$$U_1 = \frac{1}{2} \sum_{i \neq j} u_1(r_{ij}) \tag{16.17}$$

Then

$$\begin{aligned}\langle U_1\rangle_0 &= \frac{1}{Z_N^0}\int \frac{1}{2}\sum_{i\neq j} u_1(r_{ij})\exp(-\beta U_0)dr_1\cdots dr_N \\ &= \frac{N(N-1)}{2Z_N^0}\iint dr_1 dr_2 u_1(r_{12})\int\cdots\int \exp(-\beta U_0)dr_3\cdots dr_N \\ &\approx \frac{N^2}{2V^2}\iint dr_1 dr_2 u(r_{12})g(r_1,r_2) \\ &= \frac{\rho^2 V}{2}\int_0^\infty 4\pi r^2 u_1(r)g(r)dr\end{aligned} \tag{16.18}$$

where $\rho = N/V$,
$g(r)$ is the pair distribution function https://en.wikipedia.org/wiki/Radial_distribution_function:

$$g(r) = g(r_{12}) = g(r_1,r_2) = \frac{V^2(N-1)}{NZ_N}\int\cdots\int \exp(-\beta U)dr_3\cdots dr_N \tag{16.19}$$

So the free energy is

$$A = -\frac{1}{\beta}\ln\left(\frac{Z_N^0}{N!\lambda^{3N}}\right) + \frac{\rho^2 V}{2}\int_0^\infty 4\pi r^2 u_1(r)g(r)dr \tag{16.20}$$

16.2 The Van der Waals equation

Now we turn to a specific example, let $u_0(r)$ takes a particular form

$$u_0(r) = \begin{cases} 0 & r > \sigma \\ \infty & 0 < r \leq \sigma \end{cases} \tag{16.21}$$

This represents a *hard sphere* potential. In the low-density limit, the radial distribution function

$$g(r) = \exp[-\beta u_0(r)] = \begin{cases} 1 & r > \sigma \\ 0 & 0 < r \leq \sigma \end{cases} \tag{16.22}$$

$u_1(r)$ can be some arbitrary attractive potential.

Now, the first term in A^1 is

$$\begin{aligned}A^1 &= \frac{1}{2}\rho^2 V\int_0^\infty 4\pi r^2 u_1(r)g_0(r) \\ &= \frac{1}{2}\rho^2 V\int_\sigma^\infty 4\pi r^2 u_1(r)dr \\ &= -aN\rho\end{aligned} \tag{16.23}$$

where

$$a = -2\pi\int_\sigma^\infty r^2 u_1(r)dr \tag{16.24}$$

is a number dependent on σ and the specific form of $u_1(r)$.

Z_N^0 can be determined analytically. If σ=0,

$$Z_N^0 = V^N \tag{16.25}$$

However, this is not practical. If we consider two hard spheres at closest contact, we can find that the sphere has a radius σ. Hence the excluded volume for these two particles is $\frac{4}{3}\pi\sigma^3$. Let's denote this as b. Then Nb is the total excluded volume, we find that, the partition function is approximately

$$Z_N^0 = (V - Nb)^N \tag{16.26}$$

Thus,

$$A(N, V, T) \approx -\frac{1}{\beta} \ln \left[\frac{(V - Nb)^N}{N!\lambda^{3N}} \right] - \frac{aN^2}{V} \tag{16.27}$$

P can be expressed as

$$\begin{aligned} \frac{P}{kT} &= -\frac{1}{kT} \left(\frac{\partial A}{\partial V} \right)_{N,T} \\ &= \frac{N}{V - Nb} - \frac{aN^2}{kTV^2} \\ &= \frac{\rho}{1 - \rho b} - \frac{a\rho^2}{kT} \end{aligned} \tag{16.28}$$

If we let $v = N/V$,

$$P = \frac{RT}{v - b} - \frac{a}{v^2} \tag{16.29}$$

This is the famous *van de Waals equation*.

16.3 Virial expansion to 2nd order

In the very low density limit, we may assume that $\rho b \ll 1$

$$\frac{1}{1 - \rho b} \approx 1 + \rho b \tag{16.30}$$

$$\frac{P}{kT} \approx \rho + \rho^2 \left(b - \frac{a}{kT} \right) \tag{16.31}$$

From here we can also approximate the 2nd virial coefficient,

$$B_2(T) \approx b - \frac{a}{kT} = \frac{2}{3}\pi\sigma^3 + \frac{2\pi}{kT} \int_\sigma^\infty r^2 u_1(r) dr \tag{16.32}$$

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Lecture 17: Correlation and Scatterings

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17.1 Spatial correlation function

For a system with N particles, we define the *one and two number density* as follows,

$$\begin{aligned} \rho^{(1)}(r) &= \sum_{i=1}^{N} \delta(r - r_i) \\ \rho^{(2)}(r, r') &= \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \delta(r - r_i)\delta(r' - r_j) \end{aligned} \tag{17.1}$$

Clearly, the integral over all r space is simple number of atoms, and the number of atomic pairs.

$$\begin{aligned} \int \rho^{(1)}(r) dr &= N \\ \int \rho^{(2)}(r, r') dr dr' &= N(N-1) \end{aligned} \tag{17.2}$$

Higher body terms like $\rho^{(3)}$ can be obtained as well. Now we define the general correlation function $g^{(n)}$ as

$$\begin{aligned} g^{(n)}(r_1, \cdots, r_n) &= \frac{1}{\rho^n} \rho^{(n)}(r_1, \cdots, r_n) \\ &= \frac{1}{Z_N} \left(\frac{V}{N}\right)^n \frac{N!}{(N-n)!} \int dr_{n+1} \cdots dr_N e^{-\beta U(r_1, \cdots, r_N)} \end{aligned} \tag{17.3}$$

For $n = 2$, this leads to a so called radial distribution function $g^2(r_1, r_2)$,

$$\begin{aligned} g^2(r_1, r_2) &= \frac{\rho^{(2)}(r_1, r_2)}{\rho(r_1)\rho(r_2)} \\ &= \frac{1}{Z_N} \frac{V^2(N-1)}{N} \int dr_3 \cdots dr_N e^{-\beta U(r_1, \cdots, r_N)} \end{aligned} \tag{17.4}$$

For an homogeneous isotropic system, $g(r_1, r_2)$ only depends on $r_1 - r_2$, so

$$g^2(r_1, r_2) = g(|r_1 - r_2|) = \frac{\rho_N^{(2)}(r_1 - r_2)}{\rho^2} \tag{17.5}$$

This above equation used to compute $g(r)$ in numerical simulations such as molecular dynamics. So there will be two versions in which r can be either vector or scalar form. For the vector form

$$g(\boldsymbol{r}) = \frac{1}{\rho^2}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\delta(\boldsymbol{r}-\boldsymbol{r_i})\delta(\boldsymbol{r'}-\boldsymbol{r_j}) = \frac{1}{N\rho}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\delta(\boldsymbol{r}-\boldsymbol{r_i}+\boldsymbol{r_j}) \tag{17.6}$$

A more widely used case is the scalor form,

$$g(r) = \frac{1}{4\pi r^2}\frac{1}{N\rho}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\delta(r-|r_i-r_j|) \tag{17.7}$$

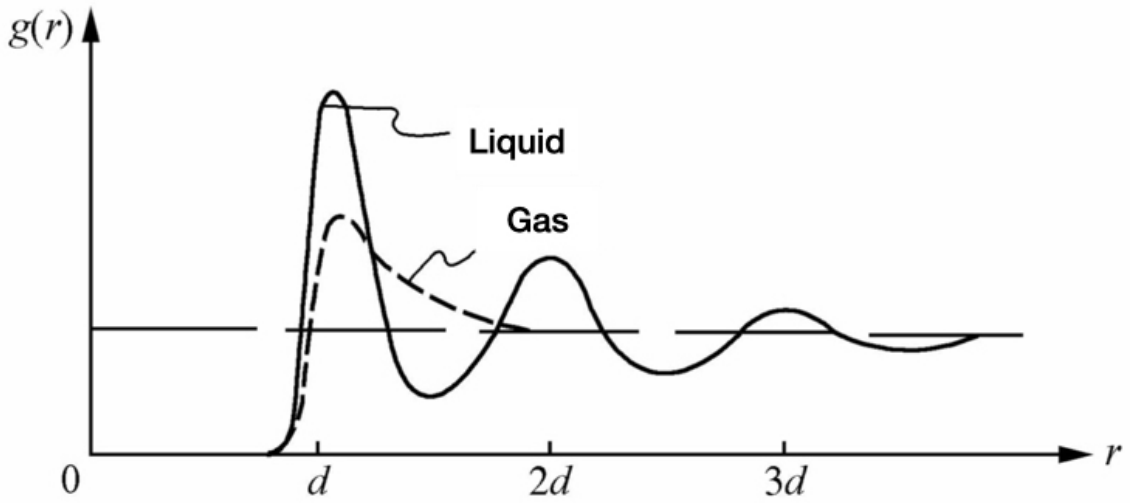
From the above equation, it is not difficult to obtain the relation that

$$\int_0^\infty 4\pi r^2 N\rho g(r)dr = N(N-1) \tag{17.8}$$

So,

$$\int_0^\infty 4\pi r^2 \rho g(r)dr = N-1 \tag{17.9}$$

Below is a typical plot of $g(r)$ for liquid and gas argon. Clearly, $g(r)$ is a spectrum with several peaks. It is always non-negative, also tends to converges to 1 in the large distances (which can be explained by its definition).



It tends out that $g(r)$ is a useful descriptor to tell the degree of order in terms of atomic packing in the 3D space. When it is ideal gas, $g(r)$ is always 1. When it is liquid, you will observe some peaks. When it is solid, the peaks will become sharper. If you integrate $g(r)$ from 0 to ∞, you get $N-1$. If you integrate from 0 to a given truncation, it tells you the number of particle within this truncation.

For dilute systems (e.g. gases), the correlations in the positions of the particles that $g(r)$ accounts for are only due to the potential $u(r)$ engendered by the reference particle, neglecting indirect effects. In the first approximation, it is thus simply given by the Boltzmann distribution law:

$$g(r) = \exp(-\beta u(r)) \tag{17.10}$$

17.2 Thermodynamic quantities

The total energy is related to the partition function

$$\begin{aligned} E &= -\frac{\partial}{\partial\beta}\ln Q \\ &= -\frac{\partial(\ln Z_N - 3N\ln\lambda - \ln N!)}{\partial\beta} \\ &= \frac{3N}{\lambda}\frac{\partial\lambda}{\partial\beta} - \frac{1}{Z_N}\frac{\partial Z_N}{\partial\beta} \\ &= \frac{3}{2}NkT + \frac{1}{Z_N}\int U(r_1,\cdots,r_N)e^{-\beta U(r_1,\cdots,r_N))}dr_1\cdots dr_N \\ &= \frac{3}{2}NkT + \langle U\rangle \end{aligned} \tag{17.11}$$

where $\lambda = (\beta h^2/2\pi m)^{1/2}$. Therefore,

In order to compute the average energy, one needs to compute the average potential U. Assume the case of a pairwise potential, so that:

$$U(r_1,\cdots,r_N) = \frac{1}{2}\sum_{i\le j} u(|r_i - r_j|) \tag{17.12}$$

$$\begin{aligned} \langle U\rangle &= \frac{1}{2Z_N}\sum_{i\le j}\int u(r_i - r_j)e^{-\beta U(r_1,\cdots,r_N)}dr_1\cdots dr_N \\ &= \frac{N(N-1)}{2Z_N}\int u(r_i - r_j)e^{-\beta U(r_1,\cdots,r_N)}dr_1\cdots dr_N \\ &= \frac{1}{2}\int u(r_1 - r_2)dr_1dr_2\left[\frac{N(N-1)}{Z_N}e^{-\beta U(r_1,\cdots,r_N)}\int dr_3\cdots dr_N\right] \\ &= \frac{1}{2}\int u(r_1 - r_2)dr_1dr_2\rho^{(2)}(r_1,r_2) \\ &= \frac{N^2}{2V^2}\int u(r_1 - r_2)g(r_1,r_2)dr_1dr_2 \\ &= \frac{N^2}{2V}\int 4\pi r^2 u(r)g(r)dr \end{aligned} \tag{17.13}$$

$$E = \frac{3}{2}NkT + \frac{N\rho}{2}\int 4\pi r^2 u(r)g(r)dr \tag{17.14}$$

$$P = -\frac{\partial A}{\partial V} = \frac{1}{\beta}\frac{\partial\ln Q}{\partial V} = \frac{1}{\beta}\frac{\partial(\ln Z_N - 3N\ln\lambda - \ln N!)}{\partial V} = \frac{1}{\beta Z_N}\frac{\partial Z_N}{\partial V} \tag{17.15}$$

We define $s_i = r_i/V^{1/3}$

$$Z_N = V^N\int ds_1\cdots ds_N e^{-\beta U(s_1V^{1/3},\cdots)} \tag{17.16}$$

$$\begin{aligned}\frac{\partial Z_N}{\partial V} &= \frac{N}{V}Z_N - \beta V^N \int ds_1 \cdots ds_N \frac{1}{3V}\sum_{i=1}^{N} r_i \frac{\partial U}{\partial r_i} e^{-\beta U(s_1 V^{1/3},\cdots,-\beta U(s_N V^{1/3})} \\ &= \frac{N}{V}Z_N - \beta \int dr_1 \cdots dr_N \frac{1}{3V}\sum_{i=1}^{N} r_i \cdot F_i e^{-\beta U(r_1,\cdots,r_N)}\end{aligned} \tag{17.17}$$

$$\frac{P}{kT} = \frac{1}{Z_N}\frac{\partial Z_N}{\partial V} = \frac{N}{V} + \frac{\beta}{3V}\langle \sum_{i=1}^{N} r_i \cdot F_i \rangle \tag{17.18}$$

$$\begin{aligned}\frac{\beta}{3V}\langle \sum_{i=1}^{N} r_i \cdot F_i \rangle &= \frac{\beta}{6V}\langle \sum_{i=1}^{N} r_{ij} \cdot F_{ij} \rangle \\ &= \frac{\beta}{6VZ_N}\int \sum_{i\le j} r_{ij} F_{ij} e^{-\beta U(r_1,\cdots,r_N)} \\ &= \frac{\beta N(N-1)}{6VZ_N}\int dr_1 \cdots dr_N r_{12} \cdot F_{12} e^{-\beta U(r_1,\cdots,r_N)} \\ &= \frac{\beta}{6V}\int dr_1 dr_2 r_{12} \cdot F_{12}\left[\frac{N(N-1)}{Z_N}\int dr_3 \cdots dr_N e^{-\beta U(r_1,\cdots,r_N)}\right] \\ &= \frac{\beta}{6V}\int dr_1 dr_2 r_{12} \cdot F_{12}\rho^{(2)}(r_1,r_2) \\ &= \frac{\beta N^2}{6V^3}\int dr_1 dr_2 r_{12} \cdot F_{12} g^{(2)}(r_1,r_2) \\ &= \frac{\beta N^2}{6V^2}\int 4\pi r^3 u'(r) g(r) dr\end{aligned} \tag{17.19}$$

$$\frac{P}{kT} = \rho - \frac{\rho^2}{6kT}\int 4\pi r^3 u'(r) g(r) dr \tag{17.20}$$

This is a general expression, especially useful to evaluate the energy of liquid and fluid.

17.3 Structure factor

The pair correlation function $g(r)$ can be measured experimentally using quasielastic scattering. If a sample is illuminated with a monochromatic beam of x-rays, neutrons, visible light, and so on, the scattered intensity as a function of the angle from the incident and its structure factor. The mathematical definition of the structure factor is:

$$S(\boldsymbol{k}) = \frac{1}{N}\langle \rho_{\boldsymbol{k}} \rho_{-\boldsymbol{k}} \rangle \tag{17.21}$$

where ρ_k is the Fourier transform of $\rho(r)$,

$$\begin{aligned} S(\boldsymbol{k}) &= \frac{1}{N}\langle \iint \exp(-i\boldsymbol{k}(r-r'))\sum_{i=1}^{N}\sum_{j=1,i\neq j}^{N}\delta(r-r_i)\delta(r'-r_j)drdr'\rangle \\ &= 1+\frac{1}{N}\iint \exp(-i\boldsymbol{k}(r-r'))\rho(r,r')drdr' \\ &= 1+\frac{\rho^2}{N}\iint \exp(-i\boldsymbol{k}(r-r'))g(r,r')drdr' \\ &= 1+\rho\int \exp(-i\boldsymbol{k}r)g(r)dr \end{aligned} \tag{17.22}$$

For isotropic fluid, we choose the scalar form,

$$\begin{aligned} S(k) &= 1+2\pi\rho\int r^2 g(r)\int_0^{\pi}\exp(-ikr\cos(\theta))\sin(\theta)d\theta dr \\ &= 1+\rho\int 4\pi r^2 g(r)\frac{\sin(kr)}{kr}dr \end{aligned} \tag{17.23}$$

This states that $S(k)$ is achieved via a Fourier transform of the pair distribution function $g(r)$. Therefore, one can use this formula to run the simulation and compare it with the experimental neutron or X-ray diffraction results.

17.4 Homework

10.4, 10.19

Physics 468/668: Statistical Physics **Spring 2019**

Lecture 18: Condensation of a Van der Waals gas

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

Various physical phenomena to which the formalism of statistical mechanics has been applied may, in general, be divided into two categories. In the first category, the microscopic constituents of the given system are, or can be regarded as, practically noninteracting, including specific heats of gases, the specific heats of solids (Section 7.4), chemical reactions and equilibrium constants (Section 6.6), the condensation of an ideal Bose gas (Sections 7.1 and 7.2). In the case of solids, the interatomic interaction does, in fact, play an important physical role; however, since the actual positions of the atoms, over a substantial range of temperatures, do not depart significantly from their mean values, we can rewrite our problem in terms of the so-called normal coordinates and treat the given solid as an assembly of practically noninteracting harmonic oscillators.

Phenomena belonging to the second category, however, present a very different situation. In most cases, one encounters analytic discontinuities or singularities in the thermodynamic functions of the given system which, in turn, correspond to the occurrence of various kinds of phase transitions. Notable examples of phenomena belonging to this category are the condensation of gases, the melting of solids, phenomena associated with the coexistence of phases (especially in the neighborhood of a critical point), the behavior of mixtures and solutions (including the onset of phase separation), the orderdisorder transitions in alloys, the transition from a normal to a superconducting material, and so on.

Mathematical problems associated with the study of cooperative phenomena are quite formidable.1 To facilitate calculations, one is forced to introduce models in which the interactions are considerably simplified, yet retain characteristics that are essential to the cooperative aspect of the problem. One then hopes that a theoretical study of these simplified models, which still involves serious difficulties of analysis, will reproduce the most basic features of the phenomena exhibited by actual physical systems.

18.1 van der Waals gas

Now we start with the following equation of state,

$$P = \frac{RT}{v-b} - \frac{a}{v^2} \tag{18.1}$$

The isotherms are shown in Figure 22.1. As shown at high temperature, the slope of $\frac{\partial P}{\partial v}$-curve is never positive. This can be understood as follows. **For a system is under equilibrium, if its volume expands, the pressure has to be reduced. Then the external environment will force the system return to the original states. Otherwise, the system is not stable.** For a rigorous mathematical proof, you can check the 2nd derivative of $\Delta^2 U > 0$ (i.e., U must be at minimum).

However, at lower temperatures there can exist a region in which the slope is zero. The existence of such regions corresponds to the coexistence of two or more phases. In other words, this indicates an onset of phase transition. Regions with positive slope should never appear. The boundary of such forbidden region can be graphically analyzed by *Maxwell construction*. The physical basis of the Maxwell construction can be seen with the help of $\mu(T, P)$. When liquid and gas reach the equilibrium,

$$\mu_{\text{liquid}} = \mu_{\text{gas}} \tag{18.2}$$

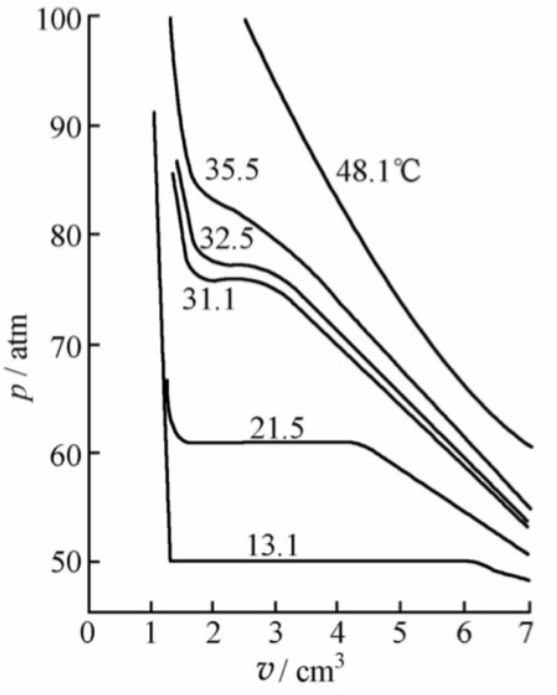


Figure 18.1: The experimentally measured CO_2 isotherms.

In the following figure, from point 1 to 3, we follow that

$$\begin{aligned} \mu_3 &= \mu_1 + \int_{P_1}^{P_3} \frac{\partial \mu}{\partial P} dP \\ &= \mu_1 + \int_{P_1}^{P_3} \frac{V}{N} dP \end{aligned} \tag{18.3}$$

To ensure that $\mu_1 = \mu_3$ from the theoretical isotherm, it require that the quantity VdP integrated along the isotherm from state 1 to state 3 much vanish. That means the area from 1-2 to 2-3 should be canceled out.

When you gradually increase the temperature from low to high, the size of "forbidden region" gradually shrinks and vanish. So there is a critical temperature T_c. It corresponds to the point $(\partial^2 P/\partial v^2)_T = 0$. Using Eq. 18.1, we obtain the coordinates of the critical point,

$$\begin{aligned} \frac{\partial P}{\partial v} &= -\frac{RT}{(v-b)^2} + \frac{2a}{v^3} = 0 \\ \frac{\partial^2 P}{\partial v^2} &= \frac{2RT}{(v-b)^3} - \frac{6a}{v^4} = 0 \end{aligned}$$

$$P_c = \frac{a}{27b^2}, \quad v_c = 3b, \quad T_c = \frac{8a}{27bR} \tag{18.4}$$

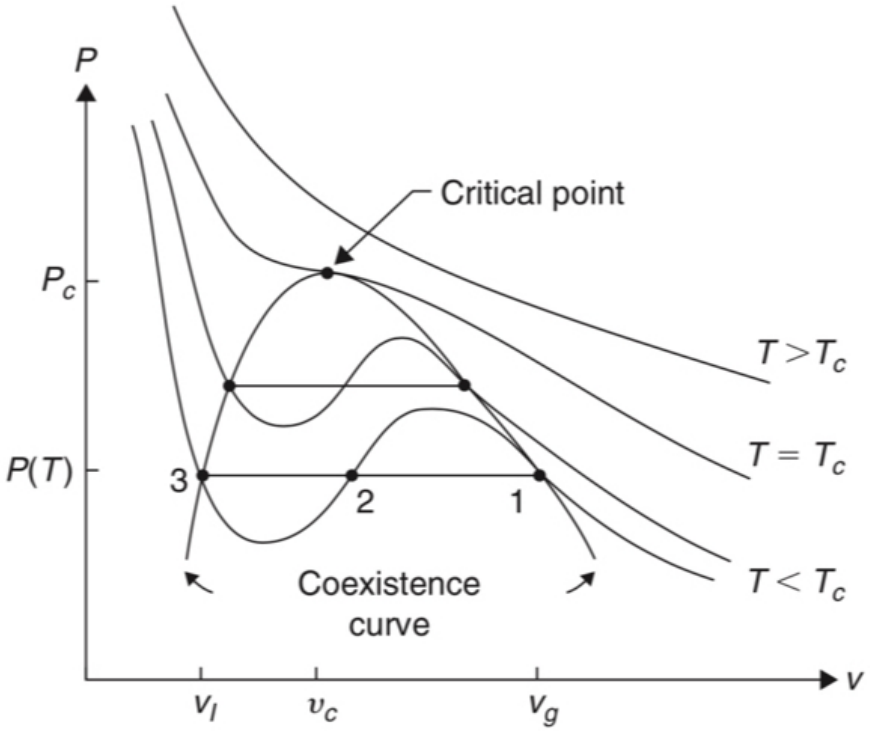


Figure 18.2: The isotherms of a van der Waals system; those for $T < T_c$ are "corrected with the help of the Maxwell construction, thus leading to the coexistence curve at the top of which sits the critical point.

We thus find that P_c, v_c, and T_c vary from system to system (through the interaction parameters a and b). However, the quantity K,

$$K = RT_c/P_c v_c = 8/3 = 2.66666 \tag{18.5}$$

has a common, universal value.

$$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T = -\frac{1}{V(\partial P/\partial V)} \tag{18.6}$$

At the critical point,

$$\begin{aligned}\frac{\partial P}{\partial V}|_{v=v_c} &= -\frac{RT}{(v_c-b)^2} + \frac{2a}{v_c^3} \\ &= -\frac{RT}{4b^2} + \frac{2a}{27b^3} \\ &= \frac{R}{4b^2}\left(\frac{8a}{27Rb} - T\right) \\ &= \frac{R}{4b^2}\left(T_c - T\right)\end{aligned} \tag{18.7}$$

Therefore,

$$\kappa_T \propto (T - T_c)^{-1} \tag{18.8}$$

$$P = P_c + \frac{\partial P}{\partial \rho}(\rho - \rho_c) + \frac{1}{2}\frac{\partial^2 P}{\partial \rho^2}(\rho - \rho_c)^2 + \cdots \tag{18.9}$$

At $P = P_c$

$$\begin{aligned} \frac{\partial P}{\partial \rho} &= \frac{\partial P}{\partial V}\frac{\partial V}{\partial \rho} = 0 \\ \frac{\partial^2 P}{\partial \rho^2} &= \frac{\partial^2 P}{\partial V^2}\left(\frac{\partial V}{\partial \rho}\right)^2 + \frac{\partial P}{\partial V}\frac{\partial^2 V}{\partial \rho^2} = 0 \end{aligned} \tag{18.10}$$

So the first nonzero term should be the third order,

$$(P - P_c) \propto (\rho - \rho_c)^{-3} \tag{18.11}$$

In general, one can prove the followings

$$C_V = (T - T_c)^{-\alpha} \quad (\alpha = 0, \text{Experiment: 0.1-0.2}) \tag{18.12}$$

$$\rho_l - \rho_g = (T - T_c)^{-\beta} \quad (\beta = 0.5, \text{Experiment: 0.32-0.35}) \tag{18.13}$$

$$\kappa = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right) = (T - T_c)^{-\gamma} \quad (\gamma = 1, \text{Experiment: 1.2-1.3}) \tag{18.14}$$

$$P - P_c = (\rho - \rho_c)^{\delta} \quad (\delta = 3, \text{Experiment: 4.6-5.0}) \tag{18.15}$$

The above four equations illustrate the nature of the critical behavior displayed by a van der Waals system undergoing the gasliquid transition. While it differs in several important respects from the critical behavior of real physical systems, it shows up again and again in studies pertaining to other critical phenomena that have apparently nothing to do with the gas-liquid phase transition.

Physics 468/668: Statistical Physics **Spring 2019**

Lecture 19: Ising model

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

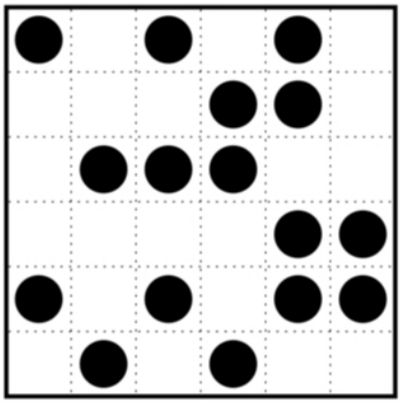


Figure 19.1: Lattice model.

19.1 A lattice Model

A number of physico-chemical systems that undergo phase transitions can be represented, to varying degrees of accuracy, by an *array of lattice sites, with only nearest-neighbor interaction that depends on the manner of occupation of the neighboring sites*. This can explain a variety of phenomena such as ferromagnetism and antiferromagnetism, gas-liquid and liquid-solid transitions, order-disorder transitions in alloys, phase separation in binary solutions, and so on.

The Hamiltonian of the system in configuration $\{\sigma_1, \sigma_2, \cdots \sigma_N\}$ is then given by

$$H\{\sigma_i\} = -J \sum_{N.N.} \sigma_i \sigma_j - \mu B \sum_i \sigma_i \tag{19.1}$$

where J is the coefficient related to two spin configurations, and each σ_i takes -1 or +1 configuration.

The partition function is

$$\begin{aligned} Q_N(B,T) &= \sum_{\sigma_1}\sum_{\sigma_2}\cdots\sum_{\sigma_N} \exp[-\beta H\{\sigma_i\}] \\ &= \sum_{\sigma_1}\sum_{\sigma_2}\cdots\sum_{\sigma_N} \exp\left[\beta J \sum_{N,N} \sigma_i\sigma_j + \beta\mu B \sum_i \sigma_i\right] \end{aligned} \tag{19.2}$$

The Helmholtz free energy, the internal energy, the specific heat, and the net magnetization of the system then follow from the formulae

$$\begin{aligned} A(B,T) &= -kT \ln Q_N(B,T) \\ U(B,T) &= -T^2 \frac{\partial}{\partial T}\left(\frac{A}{T}\right) = kT^2 \frac{\partial}{\partial T} \ln Q_N \\ C(B,T) &= \frac{\partial U}{\partial T} = -T\frac{\partial^2 A}{\partial T^2} \end{aligned} \tag{19.3}$$

and

$$\overline{M}(B,T) = \overline{\mu\left(\sum_i \sigma_i\right)} = \overline{\left(-\frac{\partial H}{\partial B}\right)} = \frac{1}{\beta}\left(\frac{\partial \ln Q_N}{\partial B}\right)_T = -\left(\frac{\partial A}{\partial B}\right)_T \tag{19.4}$$

The quantity $\bar{M}(0,T)$ gives the spontaneous magnetization of the system. If it is nonzero at temperatures below a certain critical temperature T_c, the system would be ferromagnetic for $T < T_c$ and paramagnetic for $T > T_c$. At the transition temperature itself, the system is expected to show some sort of a singular behavior.

The energy depends only on a few numbers such as the total number N_+ of up spins, the total number N_{++} of upup nearest-neighbor pairs, and so on. To see this, we define certain other numbers as well: N_- as the total number of down spins, N_{--} as the total number of downdown nearest-neighbor pairs, and N_+ as the total number of nearest-neighbor pairs with opposite spins. The numbers N_+ and N_- must satisfy the relation

$$N_+ + N_- = N \tag{19.5}$$

If q denotes the coordination number of the lattice, we have

$$\begin{aligned} qN_+ &= 2N_{++} + N_{+-} \\ qN_- &= 2N_{--} + N_{+-} \end{aligned} \tag{19.6}$$

With the help of these relations, we can express all numbers in terms of any two of them.

$$N_- = N - N_+, \quad N_{+-} = qN_+ - 2N_{++}, \quad N_{--} = \frac{1}{2}qN - qN_+ + N_{++} \tag{19.7}$$

It will be noted that the total number of pairs of all types is given,

$$N_{++} + N_{--} + N_{+-} = \frac{1}{2}qN \tag{19.8}$$

Naturally the Hamiltonian of the system can also be expressed in terms of N_+ and N_{++}; we have

$$\begin{aligned} H_N(N_+, N_{++}) &= -J(N_{++} + N_{--} - N_{+-}) - \mu B(N_+ - N_-) \\ &= -J(\frac{1}{2}qN - 2qN_+ - 4N_{++}) - \mu B(2N_+ - N) \end{aligned} \tag{19.9}$$

Now, let $g_N(N_+, N_{++})$ be "the number of distinct ways in which the N spins of the lattice can be so arranged as to yield certain preassigned values of the numbers N_+ and N_{++}. The partition function of the system

can then be written as

$$Q_N(B,T) = \sum_{N_+,N_{++}}{}' g_N(N_+,N_{++}) \exp[-\beta H_N(N_+,N_{++})] \tag{19.10}$$

So,

$$Q_N(B,T) = e^{-\beta A} = e^{-\beta N(\frac{1}{2}qJ-\mu B)} \sum_{N_+=0}^{N} e^{-2\beta(qJ-\mu B)N_+} \sum_{N_{++}}{}' g_N(N_+,N_{++}) e^{4\beta J N_{++}} \tag{19.11}$$

where the primed summation goes over all values of N_{++} that are consistent with a fixed value of N_+ and is followed by a summation over all possible values of N_+, that is from $N_+ = 0$ to $N_+ = N$. The central problem thus consists in determining $g_N(N_+, N_{++})$ for the various lattices of interest.

19.2 Lattice gas

A lattice gas is a collection of atoms, N_a in number that can occupy only discrete positions in space - positions that constitute a lattice structure with coordination number q. Each lattice site can be occupied by at most one atoom, and the interaction energy between two occupied sites is nonzero, say $-\epsilon_0$, only if the sites involved constitute a nearest neighbor pair. The configurational energy of the gas is

$$E = -\epsilon_0 N_{aa} \tag{19.12}$$

where N_{aa} is the total number of nearest-neighbor pairs in a given system. Let $g_N(N_a, N_{aa})$ denote "the number of distinct ways in which the N_a atoms of the gas, assumed indistinguishable, can be distributed among the N sites of the lattice so as to yield a certain preassigned value of N_{aa}." THe partition function of the system, neglecting the kinetic energy of the atoms, is given by

$$Q_{N_a}(N,T) = \sum_{N_{aa}}{}' g_N(N_a,N_{aa}) e^{\beta\epsilon_0 N_{aa}} \tag{19.13}$$

where the primedd summation goes over all values of N_{aa} that are consistent with the given values of N_a and N. Clearly, N here is analogical to volume available to the lattice gas.

Going over to the grand canonical ensemble, we write the paritition function

$$\mathcal{Q}(z,N,T) = \sum_{N_a=0}^{N} z^{N_a} Q N_a(N,T) \tag{19.14}$$

The pressure P and the mean number $\bar{N_a}$ of the atoms in the gas is

$$e^{\beta PN} = \sum_{N_a=0}^{N} z^{N_a} \sum_{N_a a}{}' g_N(N_a,N_{aa}) e^{\beta\epsilon_0 N_{aa}} \tag{19.15}$$

and

$$\frac{\bar{N_a}}{N} = \frac{1}{v} = \frac{z}{kT}\left(\frac{\partial P}{\partial z}\right)_T \tag{19.16}$$

The lattice gas	The ferromagnet
$N_a, N - N_a$	$N_+, N_- N_+ (= N_-)$
$-\epsilon_0$	$4J$
z	$\exp[-2\beta(qJ - \mu B)]$
P	$-\left(\frac{A}{N} + \frac{1}{2}qJ - \mu B\right)$
$\frac{\bar{N}_a}{N}$	$\frac{\bar{N}_+}{N}\left(= \frac{1}{2}\left[\frac{\bar{M}}{N\mu} + 1\right]\right)$

Table 19.1: The relation between lattice gas and binary alloy.

where

$$\bar{M} = \mu(\bar{N}_+ - \bar{N}_-) = \mu(2\bar{N}_+ - N) \tag{19.17}$$

$$\frac{\bar{N}_+}{N} = \frac{1}{kT}\left(\frac{\partial(A/N + 1/2qJ - \mu B)}{2\beta\partial(qJ - \mu B)}\right) = \frac{1}{2}\left[- \frac{1}{N\mu}\left(\frac{\partial A}{\partial B}\right)_T + 1\right] \tag{19.18}$$

which leads to

$$\frac{\bar{N}_+}{N} = \frac{1}{2}\left(\frac{\bar{M}}{N\mu} + 1\right) \tag{19.19}$$

Is this system related to any real physical system in nature? Note that this model automatically includes a short-range repulsion between the particles, since no two particles are allowed to share a same cell. The simplest instance of the lattice gas is obtained when $J = 0$, in which case the Hamiltonian is identically 0. Since its particles only interact through short-range repulsion, this model is called the hard-core lattice gas.

19.3 The Binary alloy

Much of the early activity in the theoretical analysis of the Ising model was related to the study of order-disorder transitions in alloys. In an alloy to be specific, a binary alloy we have a lattice structure consisting of two types of atoms, say 1 and 2, numbering N_1 and N_2, respectively. In a configuration characterized by the numbers N_{11}, N_{22}, and N_{12} of the three types of nearest-neighbor pairs, the configurational energy of the alloy may be written as

$$E = \epsilon_{11}N_{11} + \epsilon_{22}N_{22} + \epsilon_{12}N_{12} \tag{19.20}$$

As in the case of ferromagnet, the various numbers appearing in the expression for E may be expressed in terms of the numbers N, N_1, N_{11}, the equation can be expressed as:

$$\begin{aligned} E &= \epsilon_{11}N_{11} + \epsilon_{22}\left(\frac{1}{2}qN - qN_1 + N_{11}\right) + \epsilon_{12}(qN_1 - 2N_{11}) \\ &= \frac{1}{2}q\epsilon_{22}N + q(\epsilon_{12} - \epsilon_{22})N_1 + (\epsilon_{11} + \epsilon_{22} - 2\epsilon_{12})N_{11} \end{aligned} \tag{19.21}$$

The relation between this system and the lattice gas is now straightforward

The lattice gas	The binary alloy
$N_a, N - N_a$	$N_1, N - N_1 (= N_2)$
$-\epsilon_0$	$(\epsilon_{11} + \epsilon_{22} - 2\epsilon_{12})$
A	$A - \frac{1}{2} q \epsilon_{22} N - q(\epsilon_{12} - \epsilon_{22}) N_1$

Table 19.2: The relation between lattice gas and binary alloy.

At absolute zero, our alloy will be in the state of minimum configurational energy, which would also be the state of maximum configurational order. We expect that the two types of atoms would then occupy mutually exclusive sites, so that one might speak of atoms 1 being only at sites a and atoms 2 being only at sites b. As temperature rises, an exchange of sites results and, in the face of thermal agitation, the order in the system starts giving way. Ultimately, the two types of atoms get so mixed up that the very notion of the sites a being the right ones for atoms 1 and the sites b being the right ones for atoms 2 break down; the system then behaves, from the configurational point of view, as an assembly of $N_1 + N_2$ atoms of essentially the same species.

Physics 468/668: Statistical Physics **Spring 2019**

Lecture 20: Ising model in the zeroth approximation

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

In 1928 Gorsky attempted a statistical study of order-disorder transitions in binary alloys on the basis of the assumption that the work expended in transferring an atom from an "ordered" position to a "disordered" one is directly proportional to the degree of order prevailing in the system. This idea was further developed by Bragg and Williams (1934, 1935) who, for the first time, introduced the concept of long-range order in the sense we understand it now and, with relatively simple mathematics, obtained results that could explain the main qualitative features of the relevant experimental data. The basic assumption in the Bragg-Williams approximation was that **the energy of an individual atom in the given system is determined by the (average) degree of order prevailing in the entire system rather than by the (fluctuating) configurations of the neighboring atoms**. In this sense, the approximation is equivalent to the mean molecular field (or the internal field) theory of Weiss, which was put forward in 1907 to explain the magnetic behavior of ferromagnetic materials.

20.1 Zeroth approximation

It seems natural to call this approximation the zeroth approximation, for its features are totally insensitive to the detailed structure. So the coordination numbers will be important.

We now define a long-range order parameter L in a given configuration by the very suggestive relationship

$$L = \frac{1}{N}\sum_i \sigma_i = \frac{N_+ - N_-}{N} = 2\frac{N_+}{N} - 1 \quad (-1 \le L \le +1) \tag{20.1}$$

which gives

$$\begin{aligned} N_+ &= \frac{N}{2}(1+L) \\ N_- &= \frac{N}{2}(1-L) \end{aligned} \tag{20.2}$$

The magnetization M is then given by

$$M = (N_+ - N_-)\mu - N\mu L \quad (-N\mu \le M \le +N\mu) \tag{20.3}$$

here L is a direct measure of the net magnetization in the system. For a completely random configuration, $\overline{N_+} = \overline{N_1} = \frac{1}{2}N$, the expectation values of L and M are zero.

The expression $-J(\frac{1}{2}q\bar{\sigma})\sum_i \sigma_i$ that is, For a given σ_i, we replace each of the $q\sigma_j$ by $\bar{\sigma}$ while the factor $1/2$ is included to avoid duplication in the counting of the nearest neighbor pairs, where q is the coordination number. Making use of equation (1), and noting that $\bar{\sigma} = \bar{L}$ we obtain for the total configurational energy of the system

$$E = \sum -J\sigma_i\sigma_j - (\mu B)NL = -J(\frac{1}{2}q\bar{\sigma})\sum_i \sigma_i - (\mu B)NL = -\frac{1}{2}(qj\bar{L})NL - (\mu B)NL \tag{20.4}$$

The expectation value of E is

$$U = -\frac{1}{2}qJN\bar{L}^2 - \mu BN\bar{L} \tag{20.5}$$

In the same approximation, the difference $\delta\epsilon$ between the overall configurational energy

$$\begin{aligned} \delta\epsilon &= -J(q\bar{\sigma})\delta\sigma - \mu B\delta\sigma \\ &= 2\mu\left(\frac{qJ}{\mu}\bar{\sigma} + B\right) \end{aligned} \tag{20.6}$$

For here $\Delta\sigma = -2$. The quantity $(qJ\bar{\sigma}/\mu)$ thus stands for the internal field, which is determined by (i) long range ordering (ii) the coupling strength. The relative values of the equilibrium numbers $\bar{N_+}$ and $\bar{N_-}$ then following Boltzmann principle.

$$\frac{\overline{N_-}}{\overline{N_+}} = \exp(-\delta\epsilon/kT) = \exp[-2\mu(B' + B)/kT)] \tag{20.7}$$

Where B' denotes the internal molecular field.

$$B' = qJ\bar{\sigma}/\mu = qJ(\bar{M}/N\mu^2) \tag{20.8}$$

Therefore,

$$\frac{1-\bar{L}}{1+\bar{L}} = \exp[-2(qJ\bar{L} + \mu B)/kT] \tag{20.9}$$

or

$$\frac{qJ\bar{L} + \mu B}{kT} = \frac{1}{2}\ln\frac{1-\bar{L}}{1+\bar{L}} = \tanh^{-1}\bar{L} \tag{20.10}$$

To investigate the possibility of spontaneous magnetization, we let B $\to 0$, which leads to the relationship

$$\overline{L_0} = \tanh\left(\frac{qJ\overline{L_0}}{kT}\right) \tag{20.11}$$

This equation can be solved graphically, For any temperature T, the appropriate value of $\overline{L_0(T)}$ is determined by the point of intersection of (i) the straight line $y = L_0$ and (ii) the curve $y = \tanh(qJL_0/kT)$. Except $\overline{L_0} = 0$, we look for the nonzero solution.

Graphically, it can be seen that nonzero solution only exists when

$$qJ/kT > 1 \to T < qJ/k = T_c \tag{20.12}$$

We thus obtain a critical temperature T_c, below which the system can acquire a nonzero spontaneous magnetization and above which it cannot. Also, we see that if L_0 is a solution, $-L_0$ is the solution as well.

The precise variation of $\overline{L_0}(T)$ with T can be solved numerically. If we use the approximation $\tanh x \approx x - x^3/3$, we get

$$\overline{L_0}(T) \approx [3(1 - T/T_c)]^{1/2} \tag{20.13}$$

On the other hand, as $T \to 0$, $\overline{L_0} \to 1$ from eq. 20.11,

$$\overline{L_0}(T) \approx 1 - 2\exp(-2T_c/T) \quad (T/T_c \ll 1) \tag{20.14}$$

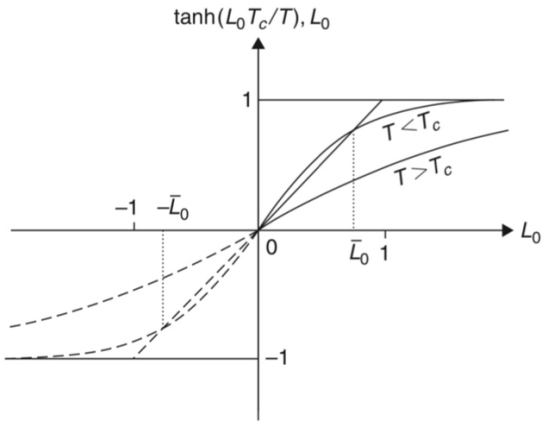


Figure 20.1: The graphical solution

The energy and specific heat are

$$U_0(T) = -\frac{1}{2}qJN\overline{L_0}^2 \tag{20.15}$$

and

$$C_0(T) = -qJN\overline{L_0}\frac{d\overline{L_0}}{dT} = \frac{Nk\overline{L_0}^2}{(T/T_c)^2/(1-\overline{L_0}^2) - T/T_c} \tag{20.16}$$

Thus, for all $T > T_c$, both $U_0(T)$ and $C_0(T)$ are zero. While at the high temperature limit ($x = TcL_0/T$),

$$C_0(T_c) = \lim_{x\to 0}\left\{\frac{Nk\cdot 3x}{\frac{(1-x)^2}{1-3x} - (1-x)}\right\} = \frac{2}{3}Nk \tag{20.17}$$

$$\begin{aligned} S_0(T_c) &= \int_0^{T_c}\frac{C_0(T)dT}{T} = -qJN\int_1^0 \frac{\overline{L_0}}{T}d\overline{L_0} \\ &= Nk\int_0^1 (\tanh^{-1}\overline{L_0})d\overline{L_0} = Nk\ln 2 \end{aligned} \tag{20.18}$$

$$\chi(B,T) = \left(\frac{\partial\bar{M}}{\partial B}\right)_T = N\mu\left(\frac{\partial\bar{L}}{\partial B}\right)_T = \frac{N\mu^2}{k}\frac{1-\bar{L}^2(B,T)}{T - T_c[1-\bar{L}^2(B,T)]} \tag{20.19}$$

For $\bar{L} \ll 1$ (which is true at high temperatures for a wide range of B but is also true near T_c if *B is small*), we obtain the Curie-Weiss law,

$$\chi_0(T) \approx (N\mu^2/k)(T - T_c)^{-1} \tag{20.20}$$

For T less than, but close to, T_c we utilize equation 20.14 as well and get

$$\chi_0(T) \approx (N\mu^2/2k)(T_c - T)^{-1} \tag{20.21}$$

Finally, using the approximation of

20.2 Remarks

Clearly there is similarity that exists between the critical behavior of a gasliquid system obeying van der Waals equation of state and that of a magnetic system treated in the Bragg-Williams approximation. Even though the two systems are physically very different, the level of approximation is such that the exponents governing power-law behavior of the various physical quantities in the critical region turn out to be the same; compare, for instance, equation (14) with (12.2.11), equations (22a) and (22b) with (12.2.13) and (12.2.14), equation (24) with (12.2.8) along with the behavior of the specific heat as well. This sort of similarity will be seen again and again whenever we employ an approach similar in spirit to the mean field approach of this section.

20.3 Random mixing approximation

The mean configurational energy in the absence of the external field is given by

$$U_0 = -J\left(\frac{1}{2}qN - 2q\overline{N}_+ 4\overline{N}_{++}\right) \tag{20.22}$$

At the same time,

$$\overline{N}_+ = \frac{1}{2}N(1+\bar{L_0}), \quad U_0 = -\frac{1}{2}qJN\bar{L_0}^2 \tag{20.23}$$

Therefore, we have

$$\bar{N}_{++} = \frac{1}{8}qN(1+\bar{L_0})^2 \tag{20.24}$$

so that

$$\frac{\bar{N}_{++}}{\frac{1}{2}qN} = \left(\frac{\bar{N}_+}{N}\right)^2 \tag{20.25}$$

Thus, the probability of having an up-up nearest neighbour pair of spins in the lattice is precisely equal to the square of the probability of having an up spin. In other words, short range ordering does not have an impact on the results.

$$\bar{N}_{++} = \frac{1}{2}qN\left(\frac{1+\bar{L}}{2}\right)^2, \quad \bar{N}_{--} = \frac{1}{2}qN\left(\frac{1-\bar{L}}{2}\right)^2 \tag{20.26}$$

$$\bar{N}_{+-} = 2*\frac{1}{2}qN\left(\frac{1+\bar{L}}{2}\right)\left(\frac{1-\bar{L}}{2}\right) \tag{20.27}$$

with the result that

$$\frac{\bar{N}_{++}\bar{N}_{--}}{(\bar{N}_{+-})^2} = \frac{1}{4} \tag{20.28}$$

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Lecture 21: Ising model in the first approximation

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

21.1 Bethe approximation

In the Bethe approximation, a given spin σ_0 is regarded a the central member of a group, which consists of this spin and it q nearest neighbors. The Hamiltonian of this system under a mean molecular field B' is

$$H_{q+1} = -\mu B\sigma_0 - \mu(B+B')\sum_{j=1}^{q}\sigma_j - J\sum_{j=1}^{q}\sigma_0\sigma_j \tag{21.1}$$

where B the external magnetic field acting on the lattice. The internal field B' is determined by the condition of self-consistency, **which requires that the mean value, $\bar{\sigma_0}$, of the central spin be the same as the mean value, $\bar{\sigma_j}$, of any of the q neighbors**.

The partition function Q of this group of spins as a whole is given by

$$\begin{aligned} Q &= \sum_{\sigma_0,\sigma_j=\pm1}\exp\left[\frac{1}{kT}\left\{\mu B\sigma_0 + \mu(B+B')\sum_{j=1}^{q}\sigma_j + J\sum_{j=1}^{q}\sigma_0\sigma_j\right\}\right] \\ &= \sum_{\sigma_0,\sigma_j=\pm1}\exp\left[\alpha\sigma_0 + (\alpha+\alpha')\sum_{j=1}^{q}\sigma_j + \gamma\sum_{j=1}^{q}\sigma_0\sigma_j\right] \end{aligned} \tag{21.2}$$

where

$$\alpha = \frac{\mu B}{kT} \quad \alpha' = \frac{\mu B'}{kT} \quad \gamma = \frac{J}{kT} \tag{21.3}$$

Now, the right side of Eq 21.1 can be written as a sum of two terms, one pertaining to $\sigma_0 = \pm 1$ and the other to $\sigma_0 = -1$, that is

$$Q = Q_+ + Q_- \tag{21.4}$$

where

$$\begin{aligned} Q_\pm &= \sum_{\sigma_j=\pm1}\exp\left[\pm\alpha + (\alpha+\alpha'\pm\gamma)\sum_{j=1}^{q}\sigma_j\right] \\ &= e^{\pm\alpha}[2\cosh(\alpha+\alpha'\pm\gamma)]^q \end{aligned} \tag{21.5}$$

The mean value of the central spin is

$$\bar{\sigma_0} = \frac{Q_+ - Q_-}{Q} \tag{21.6}$$

While the mean value of any one of its q neighbors is given by

$$\begin{aligned}\bar{\sigma}_j &= \frac{1}{q}\overline{\left(\sum_{j=1}^{q}\sigma_j\right)} = \frac{1}{q}\left(\frac{1}{Q}\frac{\partial Q}{\partial \alpha'}\right) \\ &= \frac{1}{Q}[Q_+ \tanh(\alpha+\alpha'+\gamma) + Q_- \tanh(\alpha+\alpha'-\gamma)]\end{aligned} \tag{21.7}$$

$$Q_+[1-\tanh(\alpha+\alpha'+\gamma)] = Q_-[1+\tanh(\alpha+\alpha'-\gamma)] \tag{21.8}$$

Therefore, we finally obtain

$$e^{2\alpha'} = \left[\frac{\cosh(\alpha+\alpha'+\gamma)}{\cosh(\alpha+\alpha'-\gamma)}\right]^{q-1} \tag{21.9}$$

To study the possibility of spontaneous magnetization, we get $\alpha = \mu B/kT = 0$. Equation then reduces to

$$\alpha' = \frac{q-1}{2}\ln\left\{\frac{\cosh(\alpha'+\gamma)}{\cosh(\alpha'-\gamma)}\right\} \tag{21.10}$$

If $\gamma = 0$, α' is zero. Otherwise, α' may be zero unless γ exceeds a certain critical value (γ_c). To determine this value, we do Taylor expansion around $\alpha' = 0$, which gives

$$\alpha' = (q-1)\tanh\gamma\left\{\alpha' - \operatorname{sech}^2\gamma\frac{\alpha'^3}{3}\right\} \tag{21.11}$$

For all γ, $\alpha' = 0$ is one solution (similar to what we did in the previous lecture). A non zero solution requires that,

$$(q-1)\tanh\gamma > 1 \tag{21.12}$$

That is

$$\gamma > \gamma_c = \tanh^{-1}\left(\frac{1}{q-1}\right) = \frac{1}{2}\ln\left(\frac{q}{q-2}\right) \tag{21.13}$$

This corresponding temperature is,

$$T < T_c = \frac{2J}{k}/\ln\left(\frac{q}{q-2}\right) \tag{21.14}$$

which determines the Curie temperature of the lattice. We also infer that when T is close to T_c from the left side,

$$\begin{aligned}\alpha' &\approx (3\cosh^2\gamma_c[(q-1)\tanh\gamma - 1])^{1/2} \\ &\approx [3(q-1)(\gamma-\gamma_c)]^{1/2} \\ &\approx [3(q-1)\frac{J}{kT_c}(1-\frac{T}{T_c})]^{1/2}\end{aligned} \tag{21.15}$$

The parameter $\bar{L}$ which is a measure of the long-range order in the system.

$$\bar{L} = \frac{(Q_+/Q_-)-1}{(Q_+/Q_-)+1} = \frac{\sinh(2\alpha+2\alpha')}{\cosh(2\alpha+2\alpha')+\exp(-2\gamma)} \tag{21.16}$$

In the limit $B \to 0$, when T is approaching to T_c from the left side,

$$\bar{L_0} = \frac{\sinh(2\alpha')}{\cosh(2\alpha') + \exp(-2\gamma)} \approx \frac{2\alpha'}{1 + (q-2)/q} = \frac{q}{q-1}\alpha' \tag{21.17}$$

$$\bar{L_0} \approx \left[3\frac{q}{q-1} \left\{ \frac{q}{2} \ln\left(\frac{q}{q-2} \right) \right\} \left(1 - \frac{T}{T_c} \right) \right]^{1/2} \tag{21.18}$$

We see that, when $q \gg 1$, the equation reduce to their zeroth-order approximation $[3(1 - \frac{T}{T_c})]^{1/2}$.

21.2 The correlations between spin configurations

We shall now study correlations that might exist among neighboring spins in the lattice. For this, we evaluate the numbers $\bar{N}_{++}, \bar{N}_{--}, \bar{N}_{+-}$ in terms of the parameters α, α' and γ, and compare the resulting expressions with the ones obtained under the mean field approximation. Carrying out summations in (2) over all the spins (of the group) except α_0 and σ_1, we obtain

$$Q = \sum_{\sigma_0, \sigma_1 = \pm 1} \left[\exp\{\alpha\sigma_0 + (\alpha + \alpha')\sigma_1 + \gamma\sigma_0\sigma_1\} \left\{ 2\cosh(\alpha + \alpha' + \gamma\sigma_0) \right\}^{q-1} \right] \tag{21.19}$$

Writing this as a sum of three parts pertaining, respectively, to the case (i) $\sigma_0 = \sigma_1 = \pm 1$, (ii) $\sigma_0 = -\sigma_1 = -1$, and (iii) $\sigma_0 = -\sigma_1 = \pm 1$, we have

$$Q = Q_{++} + Q_{--} + Q_{+-} \tag{21.20}$$

where,

$$\begin{aligned} \bar{N}_{++} &\propto e^{(2\alpha + \alpha' + \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} \\ \bar{N}_{--} &\propto e^{(-2\alpha - \alpha' + \gamma)} [2\cosh(\alpha + \alpha' - \gamma)]^{q-1} \\ &= e^{(-2\alpha - 3\alpha' + \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} \end{aligned} \tag{21.21}$$

and

$$\begin{aligned} \bar{N}_{+-} &\propto e^{(-\alpha' - \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} + e^{(\alpha' - \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} \\ &= 2e^{(-\alpha' - \gamma)} [2\cosh(\alpha + \alpha' + \gamma)]^{q-1} \end{aligned} \tag{21.22}$$

Normalizing these expressions with the help of the relationship

$$\bar{N}_{++} + \bar{N}_{--} + \bar{N}_{+-} = \frac{1}{2} qN \tag{21.23}$$

we obtain the desired results

$$(\bar{N}_{++}, \bar{N}_{--}, \bar{N}_{+-}) = \frac{1}{2} qN \frac{(e^{2\alpha + 2\alpha' + \gamma}, e^{-2\alpha - 2\alpha' + \gamma}, 2e^{-\gamma})}{2\{e^{\gamma}\cosh(2\alpha + 2\alpha') + e^{-\gamma}\}} \tag{21.24}$$

whereby

$$\frac{\bar{N}_{++}\bar{N}_{--}}{(\bar{N}_{+-})^2} = \frac{1}{4} e^{4\gamma} = \frac{1}{4} e^{4J/kT} \tag{21.25}$$

The last result differs significantly from the one that followed from the random mixing approximation. The difference lies in the extra factor $\exp(4J/kT)$ which, for $J > 0$, favors the formation of parallel-spin pairs $\uparrow\uparrow$ and $\downarrow\downarrow$, as opposed to antiparallel-spin pairs $\uparrow\downarrow$ and $\downarrow\uparrow$. In fact, one may regard the elementary process

$$\uparrow\uparrow + \downarrow\downarrow <=> 2\uparrow\downarrow \tag{21.26}$$

$$(\bar{N}_{++}, \bar{N}_{--}, \bar{N}_{+-})_{\bar{L}=0} = \frac{1}{2}qN\frac{(e^{\gamma}, e^{\gamma}, 2e^{-\gamma})}{4\cosh\gamma} \tag{21.27}$$

$$(\bar{N}_{++}, \bar{N}_{--}, \bar{N}_{+-})_{\bar{L}=0} = \frac{1}{2}q\frac{(1,1,2)}{4} \tag{21.28}$$

Now we evaluate the configurational energy U_0 and the specific heat C_0 of the lattice in the absence of the external field ($\alpha = 0$)

$$U_0 = -j(\frac{1}{2}qN - 2q\bar{N}_{+} + 4\bar{N}_{++})_{\alpha=0} \tag{21.29}$$

$$(\bar{N}_{+})_{\alpha=0} = \frac{1}{2}N(1+\bar{L}_0) = \frac{1}{2}N\frac{\exp(2\alpha') + \exp(-2\gamma)}{\cosh(2\alpha') + \exp(-2\gamma)} \tag{21.30}$$

$$U_0 = -\frac{1}{2}qJN\frac{\cosh(2\alpha') - \exp(-2\gamma)}{\cosh(2\alpha') + \exp(-2\gamma)} \tag{21.31}$$

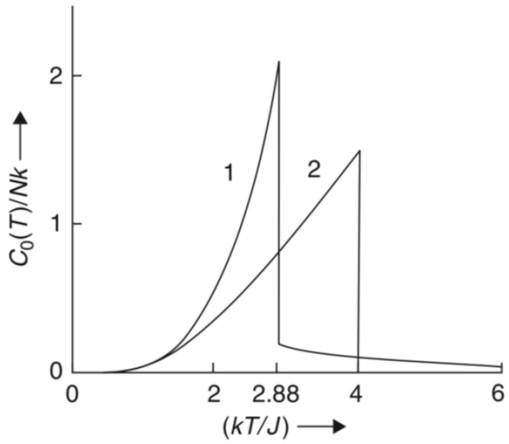


Figure 21.1: The field-free specific heat of an Ising lattice with coordination number 4. Curve 1 obtains in the Bethe approximation, curve 2 in the Bragg-Williams approximation.

$$U_0 = -\frac{1}{2}qJN\frac{1-\exp(-2\gamma)}{1+\exp(-2\gamma)} = -\frac{1}{2}qjN\tanh\gamma \tag{21.32}$$

$$C_0/Nk = \frac{1}{2}q\gamma^2\mathrm{sech}^2\gamma \quad (T > T_c) \tag{21.33}$$

$$\frac{1}{Nk}C_0(T_c+) = \frac{1}{2}q\gamma_c^2\text{sech}^2\gamma_c = \frac{1}{8}\frac{q^2(q-2)}{(q-1)^2}\left\{\ln\left(\frac{q}{q-2}\right)\right\}^2 \tag{21.34}$$

$$\begin{aligned} U_0 &= -\frac{1}{2}qJN\left[\frac{1}{q-1} + \frac{q(q-2)}{(q-1)^2}(-\gamma-\gamma_c) + \frac{q(q-2)}{(q-1)^2}\alpha'^2 + \cdots\right] \\ &= -\frac{1}{2}qJN\left[\frac{1}{q-1} + \frac{q(q-2)(3q-2)}{(q-1)^2}\frac{J}{kT_c}\left(1-\frac{T}{T_c}\right) + \cdots\right] \end{aligned} \tag{21.35}$$

$$\frac{1}{Nk}C_0(T_c-) = \frac{1}{8}\frac{q^2(q-2)(3q-2)}{(q-1)^2}\left\{\ln\left(\frac{q}{q-2}\right)\right\}^2 \tag{21.36}$$

In passing, we note that, according to Eq. 21.14, the transition temperature for a lattice with $q = 2$ is zero, which essentially means that a one-dimensional Ising chain does not undergo a phase transition. This result is in complete agreement with the one following from an exact treatment of the one-dimensional lattice; see Section 13.2. In fact, **for a lattice with $q = 2$, any results following from the Bethe approximation are completely identical with the corresponding exact results (see Problem 13.3); on the other hand, the Bragg-Williams approximation is least reliable when $q = 2$**.

That T_c for $q = 2$ is zero (rather than $2J/k$) is in line with the fact that, for all q, the first approximation yields a transition temperature closer to the correct value of T_c than does the zeroth approximation.

One important virtue of the Bethe approximation is that it brings out the role of the dimensionality of the lattice in bringing about a phase transition in the system. The fact that $T_c = 0$ for $q = 2$ and thereon it increases steadily with q leads one to infer that, while **a linear Ising chain does not undergo phase transition at any finite T, higher dimensionality does promote the phenomenon**. One may, in fact, argue that the absence of a phase transition in a one-dimensional chain is essentially due to the fact that, the interactions being severely short-ranged, communication between any two parts of the chain can be completely disrupted by a single defect in-between. The situation remains virtually unaltered even if the range of interactions is allowed to increase so long as it remains finite.

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Lecture 22: Critical exponent

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

22.1 The critical exponents

A basic problem in the theory of phase transitions is to study the behavior of a given system in the neighborhood of its critical point. We know that this behavior is marked by the fact that the various physical quantities pertaining to the system possess singularities at the critical point. It is customary to express these singularities in terms of power laws characterized by a set of critical exponents that determine the qualitative nature of the critical behavior of the given system. To begin with, we identify an order parameter m, and the corresponding ordering field h, such that, in the limit $h \to 0$, m tends to a limiting value m_0, with the property that $m_0 = 0$ for $T < T_c$ and $\pm = 0$ for $T < T_c$. For a magnetic system, the natural candidate for m is the parameter $L(= \bar{\sigma})$ of Sections 12.5 and 12.6, while h is identified with the quantity $\mu B/kT_c$; for a gasliquid system, one may adopt the density differential $(\rho_l - \rho_c)$ or $(\rho_g - \rho_c)$ for m and the pressure differential $(P - P_c)$ for h. The various critical exponents are then defined as follows.

The manner in which $m_0 \to 0$, as $T \to T_c$ from below, defines the exponent β:

$$m_0 \sim (T_c - T)^{\beta} \quad (h \to 0, T < T_c) \tag{22.1}$$

at the same time, the coefficient $q(t), r(t), s(t) \cdots$ maybe written as

$$\chi_0 \sim \left(\frac{\partial m}{\partial h}\right) \sim \begin{cases} (T - T_c)^{-\gamma} & (h \to 0, T \geq T_c) \\ (T_c - T)^{-\gamma'} & (h \to 0, T < T_c) \end{cases} \tag{22.2}$$

The equilibrium value of the order parameter is then determined by minimizing ψ_0 with respect to m_0; retaining terms only up to the order displayed in (1), which for thermodynamic stability requires that $s(t) > 0$, we obtain

$$m|_{T=T_c} \sim h^{1/\delta} \quad (T = T_c,\ h \to 0) \tag{22.3}$$

in the case of a gasliquid system, δ is a measure of the "degree of flatness of the critical isotherm at the critical point, for then

$$|P - P_c| \sim |\rho - \rho_c|^{\delta} \ \ (T = T_c, P \to P_c) \tag{22.4}$$

Finally, we define exponents α and α' on the basis of the specific heat, C_V , of the gas-liquid system:

$$C_V \sim \begin{cases} (T - T_c)^{-\alpha} & (T \geq T_c) \\ (T_c - T)^{-\alpha'} & (T \leq T_c) \end{cases} \tag{22.5}$$

In connection with the foregoing relations, especially equations (5), we wish to emphasize that in certain cases the exponent in question is rather small in value; it is then more appropriate to write

$$f(t) \sim \frac{|t|^{-\lambda} - 1}{\lambda} \quad (|t| \ll 1) \tag{22.6}$$

Table 12.1 Experimental Data on Critical Exponents

Critical Exponents	Magnetic Systems[a]	Gas–liquid Systems[b]	Binary Fluid Mixtures[c]	Binary Alloys[d]	Ferroelectric Systems[e]	Superfluid He^4[f]	Mean Field Results
α, α'	0.0–0.2	0.1–0.2	0.05–0.15	–––	–––	−0.026	0
β	0.30–0.36	0.32–0.35	0.30–0.34	0.305 ± 0.005	0.33–0.34	–––	1/2
γ	1.2–1.4	1.2–1.3	1.2–1.4	1.24 ± 0.015	1.0 ± 0.2	inaccessible	1
γ'	1.0–1.2	1.1–1.2	–––	1.23 ± 0.025	1.23 ± 0.02	inaccessible	1
δ	4.2–4.8	4.6–5.0	4.0–5.0	–––	–––	inaccessible	3
ν	0.62–0.68	–––	–––	0.65 ± 0.02	0.5–0.8	0.675	1/2
η	0.03–0.15	–––	–––	0.03–0.06	–––	–––	0

[a] Stierstadt et al. (1990).
[b] Voronel (1976); Rowlinson and Swinton (1982).
[c] Rowlinson and Swinton (1982).
[d] Als-Nielsen (1976); data pertain to beta-brass only.
[e] Kadanoff et al. (1967); Lines and Glass (1977).
[f] Ahlers (1980).

$$f(t) \sim \ln(1/|t|) \quad (|t| \ll 1) \tag{22.7}$$

For gas-liquid system obeying van der Waals equation, the various critical exponents are the same,

$$\beta = \frac{1}{2}, \;\; \gamma = \gamma' = 1, \;\; \delta = 3, \;\; \alpha = \alpha' = 0 \tag{22.8}$$

In Table 12.1 we have compiled experimental data on critical exponents pertaining to a variety of systems including the ones mentioned above; for completeness, we have included here data on another two exponents, v and η which will be defined in Section 12.12. We find that, while the observed values of an exponent, say β, differ very little as one goes from system to system within a given category (or even from category to category), these values are considerably different from the ones following from the mean field approximation. Clearly, we need a theory of phase transitions that is basically different from the mean field theory.

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Lecture 23: Landau's theory

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

23.1 A general expansion of free energy

As early as 1937 Landau attempted a unified description of all second-order phase transitions in the sense that the second derivatives of the free energy, namely the specific heat and the magnetic susceptibility (or isothermal compressibility, in the case of fluids), show a divergence while the first derivatives, namely the entropy and the magnetization (or density, in the case of fluids), are continuous at the critical point. He emphasized the importance of an order parameter m_0 (which would be zero on the high-temperature side of the transition and nonzero on the low-temperature side) and suggested that the basic features of the critical behavior of a given system may be determined by expanding its free energy in powers of m_0 (for we know that, in the close vicinity of the critical point, $m_0 \ll 1$). **He also argued that in the absence of the ordering field (h = 0) the up-down symmetry of the system would require that the proposed expansion contain only even powers of** m_0. Thus, the zero-field free energy $\psi_0 (= A_0/NkT)$ of the system may be written as

$$\psi_0(t, m_0) = q(t) + r(t)m_0^2 + s(t)m_0^4 + \cdots \quad \left(t = \frac{T - T_c}{T_c}, |t| \ll 1\right) \tag{23.1}$$

The equilibrium value of the order parameter is then determined by minimizing ψ_0 with respect to m_0; retaining terms only up to the order displayed in eq. 23.1, which for thermodynamic stability requires that $s(t) > 0$ (if we want to have ψ reach minimum), we obtain

$$r(t)m_0 + 2s(t)m_0^3 = 0 \tag{23.2}$$

The equilibrium value of m_0 is thus either 0 or $\pm\sqrt{[-r(t)/2s(t)]}$. The first solution is of lesser interest, though this is the only one we will have for $t > 0$; it is the other solutions that lead to the possibility of spontaneous magnetization in the system. To ensure that m_0 has nonzero solutions, it needs that $r(t) <0$. When $r(t) <0$, $m_0 = 0$ is no longer a stable solution, whereas it will move to other adjacent solutions.

$$|m_0| = \sqrt{-(r(t)/2s(t)} \quad (t \leq 0) \tag{23.3}$$

We see that, for $t \geq 0$, there is only one minimum, which is located at $m_0 = 0$; for $t = 0$, the minimum is rather flat. For $t < 0$, on the other hand, we have two minima, located at $m_0 = \pm m_s$, as given by expression, with a maximum at $m_0 = 0$. Now, since ψ_0 has to be convex in m_0, so that the susceptibility of the system be nonnegative. We must replace the nonconvex portion of the curve, which lies between the points $m_0 = -m_s$ and $m_0 = m_s$, by a straight line. The replacement is reminiscent of the *Maxwell construction*.

Now we can also consider the expression of $r(t)$ and $s(t)$ near T_c. Ideally, they are both very small terms which can be considered as Taylor expansion series,

$$r(t) = \sum_{k\geq 0} r_k t^k, \quad s(t) = \sum_{k\geq 0} s_k t^t, \cdots \tag{23.4}$$

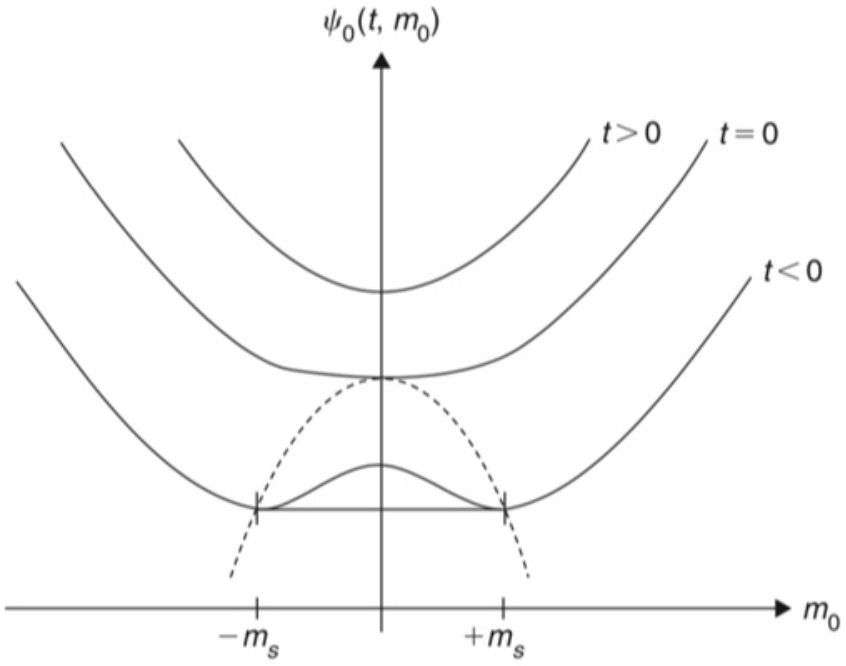


Figure 23.1: The free energy $\psi_0(t, m_0)$ of the Landau theory, shown as a function of m_0, for three different values of t. The dashed curve depicts spontaneous magnetization $m_s(t)$, while the horizontal line for $t < 0$ provides the Maxwell construction.

As you can see from Figure 23.1, at $t = 0$, it must satisfy that $r(t) = 0$ to ensure that there would be only one free energy minimum at $m = 0$. Therefore, a expansion can be

$$r(t) = r_1 t, \quad s(t) = s_0 \tag{23.5}$$

Plug it to Eq. 23.3, we get

$$|m_0| = \sqrt{-(r_1/2s_0)}|t|^{1/2} \quad (t \leq 0) \tag{23.6}$$

This gives that $\beta = 1/2$.

If $r(t)$ is a smooth function then the equilibrium value of m changes continuously from m=0 when $r(t) > 0$ to $m \neq 0$ at $r(t) < 0$. This is called 2nd order phase transition occurring at T_c with $r(T_c) = 0$.

Once we know the equilibrium value of m, we can substitute it back to the Taylor expansion

$$\psi_0(t, m_0) = \begin{cases} q(t) & t > 0 \\ q(t) - r(t)^2/2s(t) + r(t)^2/4s(t) = q(t) - r(t)^2/4s(t) = q(t) - r(t)^2/4s_0 & t < 0 \end{cases} \tag{23.7}$$

Because $r(0) = 0$, A is continuous at T_c, $S = -\partial A/\partial T$ is also continuous. But if you differentiate A twice, the term $r(t)'/2s_0$ is generally not vanishing. It means that the 2nd order derivative of A, such as C_V changes discontinuously.

23.2 Under an ordering positive field h

Now we consider there exists an ordering positive field h. If the field is weak, the only change in the expression for the free energy would be the addition of a term $-hm$.

$$\psi_h(t,m) = -hm + q(t) + r(t)m^2 + s(t)m^4 \tag{23.8}$$

The equilibrium value of m is

$$\frac{\partial \psi}{\partial m} = -h + r(t)m + 4s(t)m^3 = 0 \tag{23.9}$$

The low-field susceptibility of the system, in units of $N\mu^2/kT$, turns out to be

$$\chi = \left(\frac{\partial h}{\partial m}\right)_t^{-1} = \frac{1}{2r(t) + 12s(t)m^2} \tag{23.10}$$

valid in the limit $h \to 0$. Now for $t < 0$, $m \to 0$, we get

$$\chi \approx 1/2r_1 t \quad (t \geq 0) \tag{23.11}$$

so the critical component (γ) is 1.

For $t < 0$, $m \to \sqrt{[(r_1/2s_0)|t|]}$, we get

$$\chi \approx 1/4r_1|t| \quad (t \leq 0) \tag{23.12}$$

So γ becomes 1.

Finally, we set $t = 0$, leading to

$$h \approx 4s_0 m^3 \quad (h \to 0) \tag{23.13}$$

giving $\delta = 3$.

23.3 Heat capacity

From the thermodynamic integral, we have $dA = -SdT + PdV$ (note that PdV term in the Ising model can be considered as HdM)

$$\frac{\partial A}{\partial T} = -S, \quad \frac{\partial^2 A}{\partial T^2} = -\frac{\partial S}{\partial T} = -\frac{C}{T}, \tag{23.14}$$

In the first equation, we have $A_0/Nk = \psi_0 T$,

$$\frac{1}{Nk}\frac{\partial^2 A}{\partial T^2} = \frac{\partial^2 \psi}{\partial T^2} + 2\frac{\partial \psi}{\partial T} \tag{23.15}$$

Therefore,

$$C/T \approx -Nk\left(T_c \frac{\partial^2 \psi}{\partial t^2} + 2\frac{\partial \psi}{\partial t}\right) \tag{23.16}$$

So,

$$C \approx -NkT_c^2 \frac{\partial^2 \psi}{\partial t^2} = -NkT_c^2(2q_2 + 6q_3 t + \cdots) \approx -2q_2 NkT_c^2 \tag{23.17}$$

It means that $\alpha = 0$.

The most striking feature of the Landau theory is that it gives exactly the same critical exponents as the mean field theory or the van der Waals theory. Actually it goes much further, for it starts with an expression for the free energy of the system containing parameters $q_k, r_k, s_k, \cdots$, which represent the structure of the given system and the interactions operating in it, and goes on to show that, while the amplitudes of the various physical quantities near the critical point do depend on these parameters, the critical exponents do not!